DESIGN & LAYOUT OF CMOS BANDGAP REFERENCE USING SKY 130 PDK

A Project Phase I Report

Submitted to the APJ Abdul Kalam Technological University

in partial fulfillment of requirements for the award of degree

Bachelor of Technology

in

Electronics and Communication Engineering

by

AYSHA K H (PKD23EC024)

GOPIKA P S(PKD23EC030)

HARIKRISHNAN K S(PKD23EC034)

SHADIN P N K(PKD23EC059)

under the Guidance of

Prof. JITHESH C P

Assistant Professor, Dept of ECE



DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

GOVERNMENT ENGINEERING COLLEGE PALAKKAD

SREEKRISHNAPURAM

OCTOBER 2026

CERTIFICATE

This is to certify that the report entitled **DESIGN & LAYOUT OF CMOS BANDGAP REFERENCE USING SKY 130 PDK** submitted by **AYSHA K H - PKD23EC024**, **GOPIKA P S - PKD23EC030**, **HARIKRISHNAN K S - PKD23EC034**, **SHADIN P N K - PKD23EC059** to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in Electronics and Communication Engineering is a bonafide record of the Project Phase I carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Prof. Jithesh C P
(Project Guide)
Assistant Professor
Dept.of ECE
Government Engineering College
Palakkad

Prof. Sreeram R
(Project Coordinator)
Assistant Professor
Dept.of ECE
Government Engineering College
Palakkad

Prof. Sudheer T M
(Project Coordinator)
Assistant Professor
Dept.of ECE
Government Engineering College
Palakkad

Prof. Anwar Hussain M
Head of Department
Dept.of ECE
Government Engineering College
Palakkad

ABSTRACT

A Bandgap Reference (BGR) circuit is a fundamental building block in analog and mixed-signal integrated circuits, providing a stable, temperature-independent reference voltage by combining Complementary-to-Absolute-Temperature (CTAT) and Proportional-to-Absolute-Temperature (PTAT) voltages. It is widely used in power management ICs, ADCs, DACs, LDO regulators, sensors, and battery management systems. This project presents the design and layout of a CMOS Bandgap Reference circuit using the SKY130 (130 nm) Process Design Kit (PDK). The circuit is designed using Xschem and simulated using NGSpice to evaluate reference voltage, temperature coefficient, power consumption, line regulation, and circuit stability. The design includes a bandgap core, operational amplifier, current mirrors, bias network, and startup circuit to generate a stable reference voltage of approximately 1.2 V from a 1.8 V supply while maintaining low power consumption. The custom layout is implemented using KLayout, following good analog layout practices to minimize mismatch and parasitic effects. The layout is verified through Design Rule Check (DRC) and Layout Versus Schematic (LVS), followed by post-layout simulation to validate circuit performance. This project demonstrates the complete analog IC design flow, including schematic design, simulation, custom layout, physical verification, and post-layout validation using the open-source SKY130 CMOS technology, providing a reliable voltage reference suitable for modern analog and mixed-signal integrated circuits.

ACKNOWLEDGEMENT

First of all we thank the Almighty, for giving us the strength ,courage and knowledge to do this project successfully.

We express our heartfelt gratitude to the Principal **Dr. K R Remesh Babu**, Govt Engineering College, Sreekrishnapuram, for his inspiration throughout the project.

We would also like to thank **Prof. Anwar Hussain M**, Head Of the Department, Electronics and Communication Engineering, Govt. Engineering College, Palakkad for providing permission and availing all required facilities for undertaking the project in a systematic way.

We are extremely grateful to **Prof. Jithesh C P**, Assistant Professor, Govt. Engineering College, Palakkad, who guided us with kind, ordinal, valuable suggestions. We would like to appreciate the guidance given by **Prof. Sreeram R**, Assistant Professor and **Prof. Sudheer T M**, Assistant Professor our project Coordinators, Department of Electronics and Communication Engineering, Govt. Engineering College, Palakkad, for providing sincere guidance.

We would also like to express our sincere gratitude to all teaching and non-teaching staff of Department of Electronics and Communication Engineering, Govt. Engineering College, Palakkad, for the sincere directions imparted and the cooperation in connection with the project.

We would also like to express our heartfelt gratitude to **Soumya Gupta**, **Manjunath V K** and **Vaibhav Pavanaskar** for their constant help, technical support, and valuable contributions throughout the course of this project & to our parents for the support given in connection with the project. Gratitude may be extended to all well-wishers and our friends who supported us for the project.

Contents

Abstract **i**

Acknowledgement **ii**

List of Figures **vi**

List of Tables **viii**

List of Symbols **ix**

1 Introduction **1**

2 Literature Review **3**

3 System Overview **9**

3.1 Working 9

3.2 Block Diagram 10

3.3 Design Blocks 14

4 Startup Topologies **16**

4.1 Timing-Based Startup 16

4.2 Voltage-Comparison Startup 17

5 Bandgap Core **19**

5.1 Operating Principle 19

5.1.1 The CTAT voltage: origin and derivation 19

5.1.2 The PTAT voltage: derivation and temperature dependence 22

5.1.3 The PTAT current 24

5.1.4 Combining CTAT and PTAT for zero temperature coefficient 25

5.1.5 Forcing equal currents with feedback 25

5.1.6 Generating the reference . . . 25
5.2 Topology Selection . . . 26
5.3 Main Components and Functionality . . . 27
5.4 Advantages and Limitations . . . 28

6 Amplifier Topologies **30**
6.1 Five-Transistor OTA . . . 30
6.2 Two-Stage CMOS Op-Amp . . . 32

7 Output Topologies **35**
7.1 Current-Output Core . . . 35
7.2 Voltage-Output Core . . . 37
7.3 Regulated-Cascode Voltage Output . . . 38
7.4 Working . . . 39

8 Software Requirement **41**
8.1 XSchem . . . 41
8.2 NGSpice . . . 42
8.3 KLayout . . . 43
8.4 Skywater 130nm PDK . . . 44

9 Design Specifications **45**
9.1 Introduction . . . 45
9.2 Target Design Specifications . . . 45
9.3 Justification of Design Parameters . . . 45
9.3.1 Supply Voltage . . . 45
9.3.2 Reference Voltage . . . 46
9.3.3 Operating Temperature Range . . . 46
9.3.4 Power Consumption . . . 47
9.3.5 Startup Time . . . 47
9.4 Specification Verification . . . 47

10 Design Methodology **49**
10.1 Theoretical Background: The g_m/I_D Methodology . . . 49

10.1.1 Regions of Inversion 49
10.2 Device Characterization and Plot Analysis 50
10.2.1 NMOS Device Characterization (`nfet_01v8`) 50
10.2.2 PMOS Device Characterization (`pfet_01v8`) 53

11 Ideal Bandgap Reference Circuit Design 56
11.1 Circuit Topology 56
11.2 Design of the PTAT Voltage 57
11.2.1 Thermal Voltage Calculation 57
11.2.2 Difference Between Base-Emitter Voltages 58
11.2.3 Calculation of the PTAT Current 58
11.2.4 PTAT Scaling Resistor 59
11.3 Calculation of the Reference Voltage 59
11.4 Five-Transistor OTA Design 60
11.4.1 Tail Current and Branch Currents 61
11.4.2 Transconductance 61
11.4.3 MOSFET Aspect Ratio 62
11.4.4 OTA Voltage Gain 63
11.5 Resistor and Output-Load Design 63
11.6 Temperature Compensation 65
11.7 Simulation Results 66
11.7.1 PTAT and CTAT Characteristics 66
11.7.2 Reference Voltage Versus Temperature 67
11.7.3 Transient Response 67
11.8 Power Consumption 68
11.9 Conclusion 68

12 Conclusion 69

References 70

List of Figures

3.1 Block Diagram 10
3.2 Top-level block diagram of the CMOS Bandgap Reference 14

4.1 Timing-Based Startup Circuit 17
4.2 Voltage-Comparison Startup Circuit 18

5.1 CTAT voltage generation and its temperature dependence. 19
5.2 PTAT voltage generation and its temperature dependence. 22
5.3 Schematic of the bandgap core: (a) the basic core with equal branch currents, (b) the addition of resistors R_2 and R_3 to create constant currents, and (c) the addition of an output branch (M_3, R_L). Redrawn after the structure in [1]. 26

6.1 Schematic of the Five-Transistor OTA 31
6.2 Schematic of the Two-Stage CMOS Op-Amp 33

7.1 Current Output Core 35
7.2 Voltage-Output Core 37
7.3 Regulated-Cascode Voltage Output 39

8.1 Xschem Home Page 41
8.2 NGSpice 42
8.3 KLayout 43
8.4 Skywater 130nm PDK 44

10.1 Circuit simulation setup for NMOS (`nfet_01v8`) characterization 51
10.2 NMOS: Transconductance Efficiency (g_m/I_D) vs V_{GS} 51
10.3 NMOS: Intrinsic Gain (g_m/g_{ds}) vs g_m/I_D 52
10.4 NMOS: Transit Frequency (f_T) vs g_m/I_D 52
10.5 NMOS: Current Density (I_D/W) vs g_m/I_D 53

10.6 Circuit simulation setup for PMOS (`pfet_01v8`) characterization 53
10.7 PMOS: Transconductance Efficiency (g_m/I_D) vs V_{SG} 54
10.8 PMOS: Intrinsic Gain (g_m/g_{ds}) vs g_m/I_D 54
10.9 PMOS: Transit Frequency (f_T) vs g_m/I_D 55
10.10PMOS: Current Density (I_D/W) vs g_m/I_D 55

11.1 Schematic of the ideal bandgap reference circuit. 57
11.2 Simulated PTAT and CTAT voltage characteristics. 66
11.3 Simulated reference voltage versus temperature. 67

List of Tables

9.1 Target Design Specifications 46

11.1 Initial Design Specifications 56

List of Symbols

Ω	Unit of Resistance
V_{DD}	Supply voltage
V_{BE}	Base-emitter voltage
ΔV_{BE}	Difference in base-emitter voltages
V_T	Thermal voltage
I_S	Saturation current of the junction
k	Boltzmann's constant
q	Elementary charge / electronic charge
T	Absolute temperature in Kelvin
E_g	Silicon bandgap energy
n	Emitter-area ratio or current-density ratio
I_{PTAT}	Proportional-to-absolute-temperature current
V_{REF}/V_{out}	Reference output voltage
K	Scaling factor / gain
R_1, R_2, R_3, R_L	Resistor values
g_m	Transconductance
I_D	Drain current
g_{ds}	Drain-source conductance (output conductance)
f_T	Transit frequency
W	MOSFET channel width
L	MOSFET channel length
μ	Carrier mobility
C_{ox}	Gate-oxide capacitance per unit area
V_{OV}	Overdrive voltage
A_{OL}	Open-loop voltage gain

Chapter 1

Introduction

In modern electronic systems, stable voltage references are essential for ensuring reliable and accurate circuit operation. They are widely used in analog and mixed-signal integrated circuits, including power management integrated circuits (PMICs), low-dropout (LDO) voltage regulators, analog-to-digital converters (ADCs), digital-to-analog converters (DACs), battery management systems, sensor interfaces, and communication systems. However, variations in supply voltage, temperature, and semiconductor fabrication processes can affect circuit performance. Therefore, generating a stable reference voltage under different operating conditions is an important requirement in integrated circuit design.

A bandgap reference (BGR) circuit is an analog circuit that generates a nearly constant output voltage despite variations in supply voltage, temperature, and process parameters. A conventional bandgap reference typically produces an output voltage of approximately 1.2 V, close to the silicon bandgap voltage extrapolated to 0 K. Its operation is based on combining two voltage components with opposite temperature dependencies: a complementary-to-absolute-temperature (CTAT) voltage, which decreases with increasing temperature, and a proportional-to-absolute-temperature (PTAT) voltage, which increases with temperature. By appropriately combining these components, the temperature dependence of the output voltage can be minimized. The stability and reliability of a bandgap reference make it a fundamental building block in modern analog and mixed-signal integrated circuits.

The design of a CMOS bandgap reference involves several challenges, including temperature stability, power consumption, supply-voltage sensitivity, and reliable startup behaviour. Transistor parameter variations, component mismatch, and parasitic effects introduced during physical layout can also influence the output voltage. These challenges highlight the need for a systematic approach that considers both circuit design and physical implementation. An effective design should therefore be evaluated through circuit simulation and verified using

appropriate physical verification techniques.

The aim of this project is to design and develop the physical layout of a conventional CMOS bandgap reference circuit using the SKY130 process design kit (PDK), targeting an output reference voltage of approximately 1.2 V from a 1.8 V supply. The major objectives are to understand the operating principle of bandgap reference circuits, select a suitable conventional topology, design and simulate the required circuit, and evaluate its performance under relevant operating conditions. The work also aims to develop the physical layout, perform design rule checking (DRC) and layout-versus-schematic (LVS) verification, and consider post-layout simulation to assess the influence of layout parasitics. The overall objective is to explore a low-power and stable voltage-reference design suitable for modern integrated circuit applications.

The project follows an integrated circuit design flow using the SKY130 130 nm CMOS technology. The circuit schematic is developed using Xschem, and NGSPICE is used for circuit simulation and performance analysis. The design process involves studying the bandgap core and its CTAT and PTAT characteristics, along with the associated amplifier, current mirrors, biasing network, and startup circuit. The simulated circuit behaviour is examined to evaluate the reference voltage and its variation with temperature and supply voltage, along with other relevant performance parameters. Following schematic-level design, the physical layout is developed using KLayout. DRC and LVS checks are used to assess compliance with the process design rules and consistency between the layout and schematic. Post-layout simulation is considered to evaluate the effects of parasitic components introduced by the physical implementation.

This report presents the theoretical background, design considerations, implementation methodology, and simulation-based evaluation of the proposed bandgap reference circuit. Chapter 2 reviews relevant literature on bandgap reference circuits. Chapter 3 presents the system overview and functional blocks. Chapter 4 discusses startup topologies, while Chapter 5 explains the operating principle and topology selection of the bandgap core. Chapter 6 examines amplifier topologies, and Chapter 7 discusses output topologies. Chapter 8 describes the design tools and SKY130 technology used in the project. Chapter 9 presents the design specifications and their justification. Chapter 10 discusses the design methodology and transistor-level device characterization using the g_m/I_D methodology. Finally, Chapter 11 presents the ideal bandgap reference circuit design and simulation results, followed by Chapter 12, which concludes the report.

Chapter 2

Literature Review

Cañonero et al. (2025) proposed a 1.2 V CMOS bandgap voltage reference (BGR) for an 8-bit successive approximation register (SAR) analog-to-digital converter (ADC) intended for powerline monitoring. The design was implemented using the SKY130 130 nm CMOS process and an open-source EDA workflow involving XSCHEM and NGSpice. The proposed architecture employs a diode-connected NMOS transistor to generate complementary-to-absolute-temperature (CTAT) behaviour and a current-density ratio technique to produce a proportional-to-absolute-temperature (PTAT) voltage. A high-gain, two-stage PMOS-input operational amplifier regulates the CTAT and PTAT branches, while resistor scaling enables temperature compensation. Simulation results demonstrated temperature coefficients of 19.3 ppm/°C from 19°C to 120°C and 33 ppm/°C from 40°C to 19°C, with an average power consumption of 257.47 μW. The design achieved a power supply rejection ratio (PSRR) of 23.89 dB and a line sensitivity of 6.7%. Process corner analysis indicated reference voltages ranging from approximately 1.16 V to 1.25 V. The study demonstrated the feasibility of designing low-power CMOS bandgap references using open-source tools. However, the line sensitivity and PSRR did not meet the original design targets of below 5% and above 40 dB, respectively. These limitations highlight the need for further optimization of amplifier gain, biasing, and supply-noise rejection to improve reference stability and robustness. [2]

Razavi (2016) presented the fundamental principles and design considerations of bandgap reference circuits in the article “The Bandgap Reference: A Circuit for All Seasons.” The study explains how a temperature-independent reference voltage can be generated by combining the base-emitter voltage (V_{BE}), which exhibits a complementary-to-absolute-temperature (CTAT) characteristic, with a proportional-to-absolute-temperature (PTAT) voltage derived from the difference between the base-emitter voltages of two bipolar transistors operating at different current densities. The article discusses conventional bandgap topologies proposed by Widlar

and Brokaw, along with their implementation in CMOS technologies using bipolar devices available in standard CMOS processes. It also examines practical design challenges, including operational amplifier offset, noise, poor power supply rejection, stability under capacitive loading, and start-up conditions. Furthermore, the article presents low-voltage bandgap reference techniques that reduce the output voltage through appropriate voltage scaling while maintaining temperature compensation. The study establishes the theoretical foundation for designing stable and low-voltage bandgap reference circuits and highlights the importance of careful amplifier design, transistor sizing, and biasing to achieve reliable performance. These principles provide a useful basis for the design and optimization of CMOS bandgap reference circuits using the SKY130 process. [1]

Gunawan et al. (1993) proposed a curvature-corrected low-voltage bandgap reference circuit implemented using a standard bipolar semiconductor process with base-diffused resistors. The design combines a proportional-to-absolute-temperature (PTAT) current with a nonlinear correction current to compensate for the temperature-dependent curvature of the base-emitter voltage (V_{BE}). The generated reference current is converted into a reference voltage through a resistor, while a buffer is used to reduce the output impedance. The circuit operated with a minimum supply voltage of 1 V and achieved a reference voltage of approximately 199.2 mV after trimming, with a temperature coefficient of about ±4 ppm/°C over a temperature range of 0°C to 125°C and a supply current of approximately 100 μA. Its main contribution is the implementation of nonlinear temperature compensation to improve reference stability. However, the design relies on resistors and trimming, and resistor voltage dependence affects line regulation. Since the circuit uses a standard bipolar process, its implementation should not be described as a SKY130 CMOS design; adapting the approach to a modern CMOS process would require redesign and verification of the circuit [3].

Osaki et al. (2013) presented resistor-free bandgap reference (BGR) and sub-bandgap reference (sub-BGR) circuits for nanowatt-power CMOS integrated circuits. The proposed circuits were implemented using a 0.18-μm, 1-poly, 6-metal CMOS process with a deep N-well option. The architecture consists of a nanoampere current reference, a bipolar transistor, and proportional-to-absolute-temperature (PTAT) voltage generators based on MOSFET differential pairs operating in the subthreshold region. By eliminating conventional resistors, the design reduces power consumption and silicon area. Experimental results demonstrated a reference voltage of approximately 1.09 V at a 1.2 V supply with a power consumption of 100

nW for the BGR circuit, while the sub-BGR circuit generated approximately 0.548 V at a 0.7 V supply with a power consumption of 52.5 nW. The measured average temperature coefficients were 147 ppm/°C and 114 ppm/°C, respectively. The study also investigated process variation using 500-run Monte Carlo simulations. However, the design exhibits nonlinear temperature dependence, limited noise performance, poor output-driving capability, and a simulated startup time of 6 ms. This work is relevant to our project because it provides a theoretical and architectural foundation for bandgap reference design, particularly the use of the bipolar base-emitter voltage (V_{BE}) and PTAT voltage generation for temperature compensation. Building on these principles, our project aims to design and implement a bandgap reference circuit using the SKY130 130-nm CMOS PDK, targeting a supply voltage of 1.8 V and an output reference voltage of approximately 1.2 V. The proposed implementation will involve schematic design, pre-layout and post-layout simulation, custom layout, design rule checking (DRC), layout-versus-schematic (LVS) verification, and process, voltage, and temperature (PVT) analysis. The limitations identified in this paper motivate further investigation into temperature stability, startup behavior, and process robustness in our implementation [4].

Zhu et al. (2025) proposed a high-precision bandgap reference (BGR) circuit with low temperature drift for a multi-cell battery sensing and protection IC. The design combines a proportional-to-absolute-temperature (PTAT) current with base-current correction and high-order temperature compensation to reduce the temperature-dependent variation of the emitter-base voltage (V_{EB}). The high-order compensation circuit uses separate low-temperature and high-temperature correction currents to improve reference voltage stability over a wide temperature range. The proposed BGR achieved a temperature coefficient of approximately 5 ppm/°C. It was implemented as part of a battery management IC using a 180 nm BCD process, which supports monitoring and protection of up to 18 series-connected lithium-ion battery cells. The main contribution is the use of base-current correction and segmented curvature compensation to improve reference accuracy. However, the design uses resistors and trimming techniques, and its implementation is based on a 180 nm BCD process rather than the SKY130 CMOS process. Adapting this approach to a SKY130 PDK would require redesigning the circuit and verifying its performance through simulation and layout analysis. [5]

Huang et al. (2025) proposed a serial-stacked single-inductor dual-source triple-output energy harvesting interface with two-dimensional maximum power point tracking (2D-MPPT) for battery-free Internet of Things (IoT) devices. The design combines energy harvested from

photovoltaic (PV) and radio-frequency (RF) sources using a serial-stacked single-inductor triple-input triple-output (SS-SITITO) buck-boost converter. It employs a triple-modulation-based 2D-MPPT technique to achieve 98% maximum power point tracking accuracy, while an energy-recycling mechanism and a quad-phase buck-boost operation scheme enable a maximum output power of 26 mW. To improve energy efficiency, the circuit incorporates an edge-triggered feedback path, a power-optimized undervoltage lockout circuit, a low-voltage-supportable bandgap reference generator, and an ultra-low-power relaxation oscillator. The design was fabricated using a 180 nm CMOS process and achieved a minimum quiescent power consumption of 805 nW and a peak power conversion efficiency of 85.5%. Its main contribution is the integration of simultaneous multi-source energy harvesting, accurate MPPT, energy recycling, and multiple regulated outputs in a self-sustaining interface for battery-free IoT applications. However, the system involves multiple control circuits and requires an external inductor and storage capacitors, which may increase implementation complexity and component requirements. Although the design includes a low-voltage-supportable bandgap reference generator, its primary focus is energy harvesting and power management rather than improving the temperature stability of a standalone voltage reference. Therefore, adapting its reference-generation technique for a dedicated low-power voltage reference in a SKY130 CMOS process would require circuit redesign, implementation, and verification. [6]

Chen et al. (2021) proposed a 1.16-V curvature-compensated CMOS bandgap reference circuit with a shared offset-cancellation method for internal amplifiers. The design aims to improve reference voltage accuracy over a wide temperature range of 40 °C to 150 °C by addressing the nonlinear temperature dependence of the bipolar junction transistor (BJT) base–emitter voltage and the input-referred offset voltages of the operational amplifiers. A multi-sectional curvature-compensation technique is employed, using multiple compensation currents activated at different temperature thresholds to reduce temperature-dependent output variations. In addition, a shared auto-zeroing offset-cancellation circuit is implemented for the two internal amplifiers, reducing circuit area and power consumption compared with using separate offset-cancellation circuits. The proposed design was fabricated using a 130-nm CMOS process with a 3.3-V supply voltage. Experimental results demonstrated a measured temperature coefficient ranging from 5.8 to 13.5 ppm/°C across seven untrimmed samples, with an average of 8.75 ppm/°C. The circuit provides a nominal output voltage of approximately 1.16 V, consumes 120 A, and occupies an active area of 0.08 mm^2. The shared offset-cancellation

technique reduces amplifier-induced output errors from tens of millivolts to below 1 mV. However, the active offset-cancellation circuit increases output noise, and variations caused by resistor and BJT mismatches are not fully compensated. Furthermore, the design uses a 3.3-V supply, which may limit its direct application in low-voltage systems. Therefore, adapting this architecture to a low-voltage process such as SKY130 would require redesign and verification of the biasing, amplifier, and compensation circuits to satisfy the required supply-voltage, power-consumption, and temperature-stability specifications. [7]

Ming et al. (2018) proposed a curvature-compensated resistor-less bandgap reference circuit based on successive voltage-step compensation. The design generates successive ΔV_{GS} voltages with different temperature characteristics to cancel the thermal nonlinearity of the diode step-by-step, achieving high-order compensation without the large chip area typically required by high-resistance components. Additionally, a supply noise bypassing technique (SNBT) is utilized to ensure a high power supply rejection (PSR) performance. Fabricated in a 0.5-μm CMOS process, the circuit operates at a typical 3.6-V supply and generates a 1.196 V reference. Experimental results demonstrated a temperature coefficient of 3.98 ppm/°C and a power-supply noise attenuation of -84 dB at 100 Hz, while consuming a quiescent current of 38 μA. The main contribution is the elimination of resistors while maintaining high-order temperature stability and high PSR. However, the design requires a two-step trimming process to achieve its optimal temperature coefficient and dictates a minimum supply voltage of 2.1 V, which exceeds the 1.8 V target of standard low-voltage implementations. While the resistor-less approach saves area, adapting this architecture to the SKY130 process would require a careful redesign of the subthreshold compensation stages to account for specific process threshold variations. [8]

Hunter and Matthews (2017) developed a precision single-trim switched-capacitor bandgap reference designed specifically for battery monitoring applications. Implemented in a 0.18-μm BiCMOS process, the architecture utilizes fully differential sampling to reduce errors caused by channel charge injection and clock feedthrough, which frequently degrade pseudo-differential sampling circuits. A novel technique is introduced to sample the base-emitter voltage (V_{be}) directly onto the output of the differential amplifier, eliminating the error usually introduced when sampling the V_{be} and ΔV_{be} terms independently. The reference provides a discrete-time 3.65 V output to a delta-sigma ADC and achieves an excellent temperature drift of ±3 ppm/°C over a range of -40°C to 110°C with a single room-temperature trim. While the

design offers superior drift performance and robustness against switching errors, its switched-capacitor nature means it provides a discrete-time reference rather than a continuous-time voltage. Additionally, the system requires a 5.2 V supply, consumes a relatively high current of 750 μA during active conversion, and relies on a BiCMOS technology node. Because our SKY130 project targets a continuous-time 1.2 V reference using a 1.8 V supply in a pure CMOS process, this specific discrete-time architecture cannot be directly ported; however, the theoretical insights into single-trim precision and offset error reduction remain highly relevant for optimizing continuous-time error amplifiers. [9]

Chapter 3

System Overview

3.1 Working

A Bandgap Reference (BGR) is an analog circuit that generates a nearly constant reference voltage despite variations in temperature, supply voltage, and process parameters. In this project, a current-mode CMOS Bandgap Reference is designed using the SKY130 130-nm CMOS Process Design Kit (PDK). The design aims to generate a reference voltage of approximately 1.2 V using a 1.8 V supply.

The basic operating principle of a BGR is to combine two temperature-dependent components: a Complementary-to-Absolute-Temperature (CTAT) component and a Proportional-to-Absolute-Temperature (PTAT) component. These components exhibit opposite temperature characteristics. The CTAT component decreases with increasing temperature, whereas the PTAT component increases with temperature. By appropriately combining them, the overall temperature dependence of the output can be significantly reduced.

The proposed current-mode architecture uses bipolar junction transistors (BJTs), resistors, MOSFET-based current mirrors, and biasing circuitry to generate and combine the required currents. The BJT base-emitter voltage provides the CTAT component, while the difference between the base-emitter voltages of two BJTs operating at different current densities provides the PTAT component.

The CTAT voltage is converted into a current using a resistor, producing a current that generally decreases with increasing temperature. Similarly, the PTAT voltage difference is converted into a current using another resistor, producing a current that increases with temperature. Current mirrors copy and scale these currents so that their relative contributions can be adjusted.

The weighted PTAT and CTAT currents are then combined to generate a reference current.

Passing this current through an appropriate resistor produces the output reference voltage. The resistor values and current-mirror ratios are selected so that the positive temperature dependence of the PTAT contribution compensates for the negative temperature dependence of the CTAT contribution.

The circuit is designed and simulated using Xschem and NGSpice with SKY130 device models. After satisfactory schematic-level performance is obtained, the circuit layout is implemented using a physical design tool such as KLayout or Magic. Design Rule Check (DRC), Layout Versus Schematic (LVS), and parasitic extraction are then performed to verify the physical implementation and evaluate post-layout performance.

3.2 Block Diagram

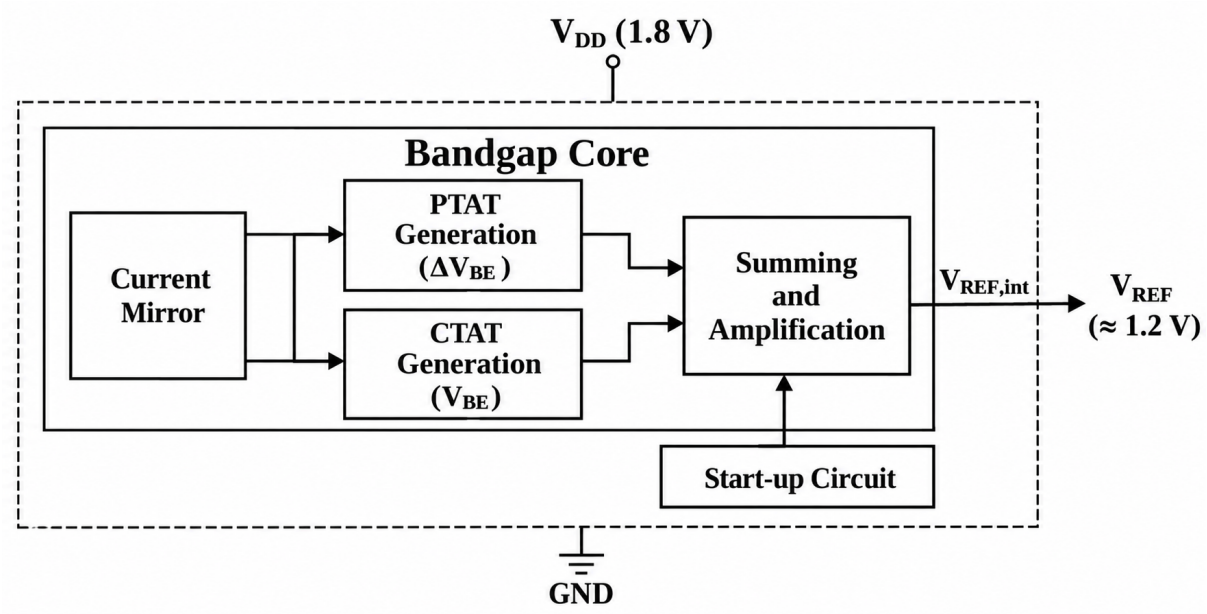


Figure 3.1: Block Diagram

The block diagram illustrates the architecture of a CMOS Bandgap Reference (BGR) circuit designed using the SKY130 130-nm CMOS Process Design Kit (PDK). The primary objective of the circuit is to generate a stable reference voltage of approximately 1.2 V from a 1.8 V supply, with minimal variation due to temperature changes. The circuit achieves temperature compensation by combining a Complementary-to-Absolute-Temperature (CTAT) voltage with a Proportional-to-Absolute-Temperature (PTAT) voltage. The project focuses on designing and analyzing the BGR circuit and implementing its physical layout using the SKY130 technology.

The major blocks of the circuit are the current mirror, PTAT generation, CTAT generation, summing and amplification, and start-up circuit. Each block performs a specific function in generating a stable reference voltage.

1. Current Mirror

The current mirror provides the required bias currents for the PTAT and CTAT generation blocks. It copies or scales a reference current to establish suitable operating conditions for the circuit. Proper biasing ensures that the transistor devices operate in their intended regions and that the PTAT and CTAT components are generated reliably.The current mirror is an essential component because the performance of the bandgap reference depends on accurate biasing and the appropriate relationship between the currents in different branches.

2. PTAT Voltage Generation

The Proportional-to-Absolute-Temperature (PTAT) generation block produces a voltage that increases approximately linearly with absolute temperature. It uses the difference in base-emitter voltages, ΔV_{BE}, of two bipolar junction transistors (BJTs) operating at different current densities. The PTAT voltage is expressed as

$$V_{\text{PTAT}} = \Delta V_{BE}$$

where:

$$\Delta V_{BE} = V_{BE1} - V_{BE2}$$

For BJTs operating at different current densities, the difference in base-emitter voltage is given by

$$\Delta V_{BE} = \frac{kT}{q} ln(n)$$

- k is Boltzmann's constant.
- T is the absolute temperature in kelvin.
- q is the electronic charge.
- n is the ratio of the effective current densities, determined by the transistor current ratio and emitter-area ratio.

Since ΔV_{BE} is proportional to absolute temperature, the PTAT voltage increases as temperature rises. This positive temperature dependence is used to compensate for the negative temperature dependence of the CTAT voltage.

3. CTAT Voltage Generation

The Complementary-to-Absolute-Temperature (CTAT) generation block produces a voltage that decreases as temperature increases. In a conventional bandgap reference, this voltage is obtained from the base-emitter voltage, V_{BE}, of a BJT.

The CTAT voltage can be expressed as

$$V_{\text{CTAT}} = V_{BE}$$

The base-emitter voltage decreases with increasing temperature under the appropriate bias conditions. Therefore, the CTAT component exhibits a negative temperature coefficient.The CTAT voltage provides the principal voltage component, while the PTAT voltage compensates for its temperature-dependent variation. Combining these two components with suitable scaling allows the circuit to maintain a nearly constant output voltage over a wide temperature range.

4. Summing and Amplification Block

The summing and amplification block combines the PTAT and CTAT components to generate the final reference voltage. Since the two components have opposite temperature coefficients, their contributions can be adjusted to minimize the overall temperature dependence.

For a conventional voltage-mode bandgap reference, the output voltage is expressed as

$$V_{\text{REF}} = V_{BE} + K\Delta V_{BE}$$

where:

- V_{REF} is the reference output voltage.
- V_{BE} is the CTAT component.
- ΔV_{BE} is the PTAT component.
- K is a scaling factor used to adjust the PTAT contribution.

The scaling factor is selected so that the positive temperature coefficient of the PTAT component approximately cancels the negative temperature coefficient of the CTAT component.Consequently, the output voltage remains nearly constant with temperature. For this

project, the target reference voltage is approximately 1.2 V, obtained using a supply voltage of 1.8 V.

5. Start-up Circuit

The start-up circuit ensures that the bandgap reference reaches its intended operating condition when the supply voltage is applied.Some bandgap reference circuits can settle into an undesired zero-current operating state during power-up. In this condition, the current mirror and voltage-generation blocks may fail to establish the required bias currents.The start-up circuit temporarily injects or establishes the necessary bias to move the circuit away from this undesired state. Once normal operation is established, the start-up circuit becomes inactive or has minimal influence on the steady-state operation.This block is important for achieving reliable circuit activation and predictable start-up behavior. The project also identifies start-up time and transient response as important performance parameters.

6. Power Supply and Ground

The circuit operates with a nominal supply voltage of

$$V_{\mathrm{DD}} = 1.8\,\mathrm{V}$$

The supply provides the energy required for the current mirror, voltage-generation blocks, and summing and amplification circuitry.Ground (GND) acts as the common voltage reference for the circuit. Proper power-supply and ground connections are necessary for correct biasing and stable operation.The design uses the SKY130 130-nm CMOS technology, which supports the implementation of the circuit using CMOS devices and the associated design and verification tools.

7. Reference Voltage Output

The output of the summing and amplification block is the reference voltage, V_{REF}. The target output is approximately 1.2 V.The main characteristic of a bandgap reference is that its output voltage should exhibit minimal variation with temperature. This is achieved by appropriately combining the CTAT and PTAT components.The reference voltage can be used as a stable voltage source in analog and mixed-signal systems, including analog-to-digital converters (ADCs), digital-to-analog converters (DACs), low-dropout regulators (LDOs), and power-management circuits.The project aims to evaluate the reference voltage, temperature coefficient, power consumption, transient response, and robustness against process, voltage, and temperature (PVT) variations.

3.3 Design Blocks

The complete bandgap reference system is divided into four primary functional blocks. Figure 3.2 illustrates the top-level signal flow connecting these stages to generate a stable reference voltage.

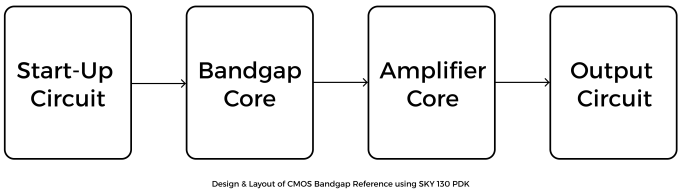


Figure 3.2: Top-level block diagram of the CMOS Bandgap Reference

Start-Up Circuit Self-biased bandgap reference circuits contain a stable ”zero-current” state. If the circuit powers up into this degenerate state, it will remain off indefinitely. The start-up circuit actively monitors the system at power-on to detect if the core is inactive. If the output or internal node voltages are too low, the start-up circuit injects an initial current into the core to force the transistors into their active regions. Once the bandgap core establishes its nominal operating currents and reaches steady state, the start-up circuit automatically disengages to prevent interference with the precision reference.

Bandgap Core The bandgap core is the central temperature-sensing engine of the design. It utilizes bipolar junction transistors (BJTs) with different emitter areas to generate a base-emitter voltage (V_{BE}) and a difference in base-emitter voltages (ΔV_{BE}). The V_{BE} provides a Complementary-to-Absolute-Temperature (CTAT) profile, while the ΔV_{BE} generates a Proportional-to-Absolute-Temperature (PTAT) current across a resistor network. The core integrates these opposing thermal trends using precisely sized resistors (R_1, R_2, R_3) to synthesize a branch current that is approximately independent of temperature variations.

Amplifier Core The amplifier core is an operational transconductance amplifier (OTA) that enforces the necessary voltage constraints within the bandgap core. It senses the voltages at the two critical branches of the BJT network (V_X and V_Y) and amplifies any difference between them. The amplifier’s output drives the common gate terminal of the core’s PMOS current mirrors. By applying negative feedback, the amplifier continuously adjusts the PMOS gate voltage until $V_X \approx V_Y$, ensuring that the PTAT and CTAT currents are generated accurately.

Output Circuit The output circuit converts the temperature-compensated current synthesized in the bandgap core into the final usable reference voltage. It typically utilizes a PMOS current mirror to copy the stable current from the core into an isolated output branch. This current is then driven through a precise load resistor (R_L). By Ohm's law, the constant current passing through the load resistor establishes the final, stable 1.2 V output voltage

Chapter 4

Startup Topologies

A startup circuit is used in a bandgap reference (BGR) to ensure that the circuit reaches its intended operating condition when the power supply is switched on. The main BGR core may fail to start automatically if it settles at an undesired operating point. Startup topologies overcome this problem by temporarily providing the necessary initial condition to activate the main circuit. Two startup approaches are discussed below: timing-based startup and voltage-comparison startup.

4.1 Timing-Based Startup

The timing-based startup technique temporarily pulls node P low during power-up to initiate the operation of the main BGR core. It uses an RC timing network to generate a time-dependent voltage that controls the startup transistor.

Main Components

The timing-based startup circuit consists of a resistor R_S and a capacitor C_S, which together generate a time-dependent voltage at node G. The startup MOSFET M_S draws current from node P when activated. Node P controls the main PMOS transistors in the BGR core and plays an important role in establishing the required core currents.

Working Principle

When the supply voltage V_{DD} is switched on, its rise causes the voltage at node G to change through the RC timing network. Once the voltage V_G reaches the required level, the startup transistor M_S turns on. The transistor then draws current from node P, pulling it low and helping the main PMOS devices turn on to establish the required core currents. After the supply voltage stabilizes, the voltage at the timing node returns towards its inactive state, causing M_S to switch off. The startup assistance is thereby removed, and the main BGR feedback loops

continue to maintain the reference voltage during normal operation.

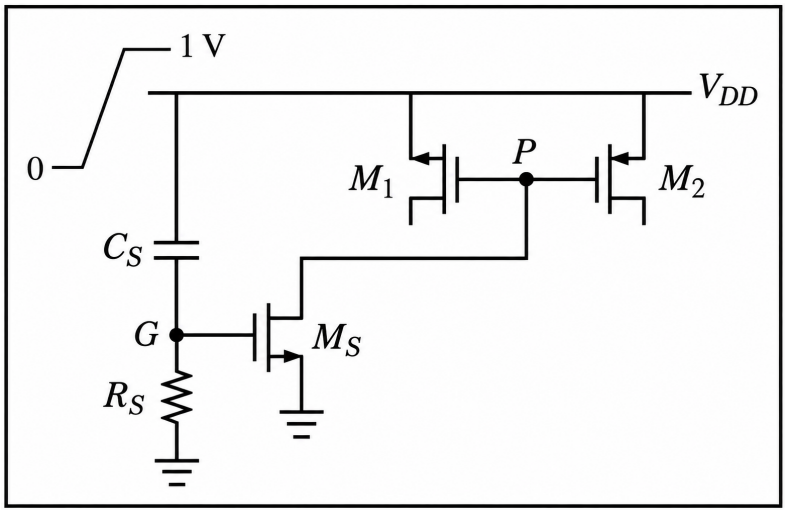


Figure 4.1: Timing-Based Startup Circuit

Advantages

The timing-based startup technique is conceptually simple and does not require an additional amplifier for comparing the output voltage against a reference. Furthermore, an RC network can generate the startup pulse required to activate the main BGR core.

Limitations

The timing network must operate correctly under different supply-voltage ramp rates, process variations, and temperature conditions. In particular, a very slow supply ramp can require extremely large timing resistance and capacitance values. The paper gives a supply-ramp example of 1 ms, for which the required timing components may become impractically large.

4.2 Voltage-Comparison Startup

The voltage-comparison startup technique determines whether the BGR circuit has reached its intended operating condition by monitoring its actual output voltage. Unlike the timing-based approach, which estimates when startup should finish, this technique uses an output-voltage comparison to determine when startup assistance is required and when it can be removed.

Main Components

The circuit consists of a startup operational transconductance amplifier (OTA) or compara-

tor, denoted by A_3, which monitors the BGR output voltage V_{out} against a startup comparison reference voltage V_r. It also includes a startup transistor M_S, which draws current from node P to help activate the main PMOS transistors in the BGR core.

Working Principle

At power-up, the output voltage V_{out} may initially be zero because the main BGR core has not yet started operating. The startup amplifier A_3 compares the actual output voltage with the reference voltage V_r to determine whether the output is sufficiently low. If the output voltage is too low, A_3 activates the startup transistor M_S. The transistor draws current from node P, helping the main BGR core turn on and establish its normal operating condition. As the core becomes operational, the output voltage rises towards its intended value. Once the comparison no longer indicates an insufficient output voltage, the amplifier disables the startup action, and the circuit continues normal operation without startup assistance.

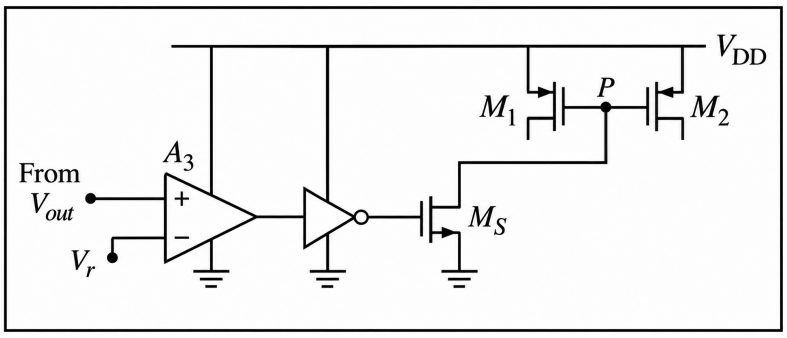


Figure 4.2: Voltage-Comparison Startup Circuit

Advantages

The voltage-comparison startup technique responds to the actual output condition rather than relying solely on an RC delay. This allows it to accommodate variations in startup time and avoids the need for a very large timing network designed to handle slow supply-voltage ramps.

Limitations

This technique requires an additional amplifier and startup transistor, increasing circuit complexity. The startup threshold is affected by the comparison reference voltage and amplifier offset. Furthermore, the startup circuit must disengage correctly once the BGR core reaches its normal operating condition.

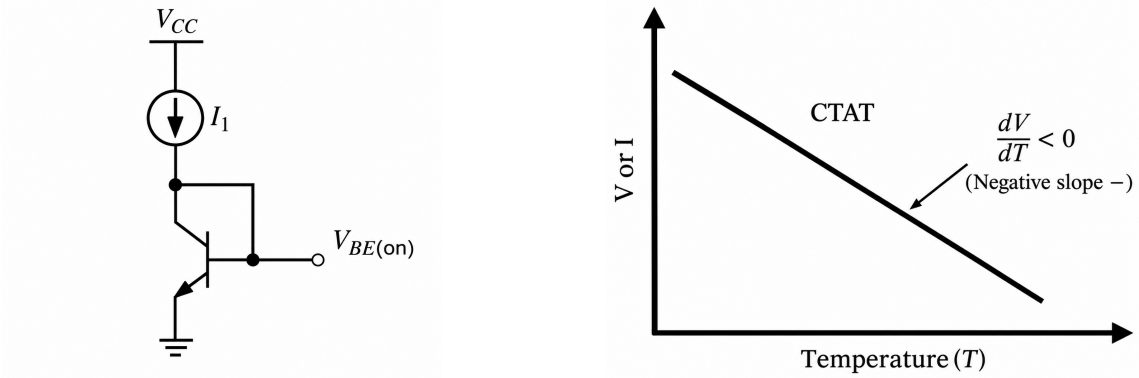
Chapter 5

Bandgap Core

The core of the bandgap reference (BGR) is the circuit that generates a voltage or current with a well-defined temperature behaviour. It combines two quantities with opposite temperature coefficients: a voltage that is *complementary to absolute temperature* (CTAT), taken from the base–emitter voltage of a bipolar transistor, and a voltage that is *proportional to absolute temperature* (PTAT), obtained from the *difference* of two base–emitter voltages. The op-amp studied in Chapter 4 closes the feedback loop around this core.

5.1 Operating Principle

5.1.1 The CTAT voltage: origin and derivation



(a) CTAT generation circuit

(b) CTAT voltage versus temperature

Figure 5.1: CTAT voltage generation and its temperature dependence.

The base-emitter voltage, V_{BE}, of a bipolar junction transistor (BJT) exhibits a complementary-to-absolute-temperature (CTAT) characteristic. At approximately constant collector current, the base-emitter voltage decreases as temperature increases, typically with a temperature coefficient of about $-2\,\mathrm{mV/°C}$. The base–emitter voltage of a diode-connected bipolar

transistor carrying a bias current I_0 is

$$|V_{BE}| = V_T \ln\left(\frac{I_0}{I_S}\right), \qquad V_T = \frac{kT}{q}, \tag{5.1}$$

where I_S is the saturation (scale) current of the junction. Two factors depend on temperature, and they act in opposite directions:

- V_T is directly proportional to T, so it is PTAT, with

$$\frac{\partial V_T}{\partial T} = \frac{k}{q} \approx 0.086 \text{ mV/K}. \tag{5.2}$$

- $\ln(I_0/I_S)$ is CTAT, because I_S increases very rapidly with temperature. Here the bias current I_0 is assumed to be constant with temperature, $\partial I_0/\partial T = 0$.

Which of the two wins is found by differentiating Eq. (5.1).

Step 1: Temperature dependence of I_S. The saturation current is proportional to the carrier mobility μ, to kT and to the square of the intrinsic carrier concentration n_i:

$$I_S \propto \mu\, kT\, n_i^2. \tag{5.3}$$

The mobility and the intrinsic concentration vary with temperature as

$$\mu \propto \mu_0 T^m \;\; (m \approx -\tfrac{3}{2}), \qquad n_i^2 \propto T^3 \exp\left(-\frac{E_g}{kT}\right), \tag{5.4}$$

where E_g is the silicon bandgap energy. Combining the three terms gives

$$I_S = b\, T^{4+m} \exp\left(-\frac{E_g}{kT}\right), \tag{5.5}$$

with b a temperature-independent constant. The exponential term makes I_S roughly double or more for every few degrees of temperature rise.

Step 2: Derivative of I_S. Differentiating Eq. (5.5) with the product rule,

$$\begin{aligned}\frac{\partial I_S}{\partial T} &= b\left[T^{4+m}\exp\!\left(-\frac{E_g}{kT}\right)\frac{E_g}{kT^2} + (4+m)\,T^{3+m}\exp\!\left(-\frac{E_g}{kT}\right)\right] \\ &= I_S\left[\frac{4+m}{T} + \frac{E_g}{kT^2}\right].\end{aligned} \tag{5.6}$$

The first term comes from the power of T and the second from the exponential.

Step 3: Derivative of V_{BE}. Applying the product rule to Eq. (5.1) with $\partial I_0/\partial T = 0$,

$$\frac{\partial V_{BE}}{\partial T} = \frac{\partial V_T}{\partial T}\ln\!\left(\frac{I_0}{I_S}\right) - \frac{V_T}{I_S}\frac{\partial I_S}{\partial T}. \tag{5.7}$$

Substituting Eq. (5.2) and Eq. (5.6), the factor I_S cancels:

$$\frac{\partial V_{BE}}{\partial T} = \frac{k}{q}\ln\!\left(\frac{I_0}{I_S}\right) - V_T\left[\frac{4+m}{T} + \frac{E_g}{kT^2}\right]. \tag{5.8}$$

Now $\dfrac{k}{q}\ln\!\left(\dfrac{I_0}{I_S}\right) = \dfrac{V_T}{T}\ln\!\left(\dfrac{I_0}{I_S}\right) = \dfrac{V_{BE}}{T}$, and $V_T\dfrac{E_g}{kT^2} = \dfrac{E_g}{qT}$. Therefore

$$\boxed{\frac{\partial V_{BE}}{\partial T} = \frac{V_{BE} - (4+m)\,V_T - E_g/q}{T}} \tag{5.9}$$

Step 4: Numerical value. For $V_{BE} \approx 0.7\,\text{V}$, $m = -3/2$, $V_T = 26\,\text{mV}$, $E_g/q \approx 1.2\,\text{V}$ and $T = 300\,\text{K}$, the term $(4+m)V_T = 2.5 \times 26\ \text{mV} = 65\,\text{mV}$, so

$$\frac{\partial V_{BE}}{\partial T} \approx \frac{0.7 - 0.065 - 1.2}{300} \approx -1.9\ \text{mV/K}. \tag{5.10}$$

Using the more accurate silicon value $E_g/q \approx 1.12\,\text{V}$ gives $\approx -1.6\,\text{mV/K}$, in agreement with the textbook value. The slope is therefore about -1.5 to $-2\,\text{mV/K}$, depending on the assumptions, and it is negative: V_{BE} is CTAT.

Discussion.

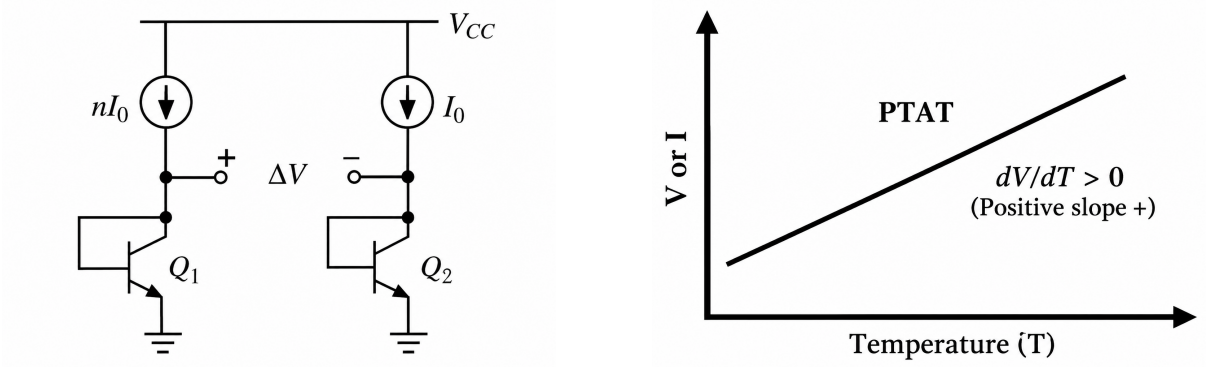
- Eq. (5.9) shows that the slope depends on V_{BE} itself: a larger V_{BE} (e.g. 750 mV instead of 700 mV) gives a less negative slope ($\approx -1.45\,\text{mV/K}$ for $E_g/q = 1.12\,\text{V}$).
- The temperature dependence of V_{BE} is not exactly linear. The $(4+m)V_T$ term and the

$1/T$ factor in Eq. (5.9) cause a small curvature, which limits the accuracy of a first-order bandgap reference.

- The bias current matters. In the derivation I_0 is constant. In this core the branch current is PTAT, $I_D = V_T \ln n/R_1$, so $\ln(I_0)$ adds a small positive term of about $+V_T/T = +0.086\,\text{mV/K}$ to the slope.
- At room temperature $|V_{BE}| \approx 750\,\text{mV}$ for the transistors used here.

5.1.2 The PTAT voltage: derivation and temperature dependence

The PTAT voltage is obtained from the *difference* of the base–emitter voltages of two bipolar transistors Q_1 and Q_2 that have different emitter areas. The derivation starts from the diode equation and follows five steps.



(a) PTAT generation circuit

(b) PTAT voltage versus temperature

Figure 5.2: PTAT voltage generation and its temperature dependence.

Step 1: Base–emitter voltage of each transistor. For a diode-connected BJT in the forward-active region,

$$I_C = I_S \left(e^{V_{BE}/V_T} - 1\right) \approx I_S\, e^{V_{BE}/V_T}, \tag{5.11}$$

since $V_{BE} \gg V_T$ in normal bandgap operation. Solving for V_{BE},

$$V_{BE1} = V_T \ln\!\left(\frac{I_{C1}}{I_{S1}}\right), \qquad V_{BE2} = V_T \ln\!\left(\frac{I_{C2}}{I_{S2}}\right). \tag{5.12}$$

Step 2: Saturation currents and the emitter-area ratio. The saturation current is proportional to the emitter area A_E,

$$I_{S1} = A_{E1} J_S(T), \qquad I_{S2} = A_{E2} J_S(T), \tag{5.13}$$

where $J_S(T)$ is the saturation current density. If the two devices use the same technology and layout style, $J_S(T)$ is identical, and

$$\frac{I_{S2}}{I_{S1}} = \frac{A_{E2}}{A_{E1}} \equiv n. \tag{5.14}$$

In this design Q_1 is made of four unit devices and Q_2 of sixty-four, so $n = 16$.

Step 3: The difference.

$$\begin{aligned} \Delta V_{BE} &= V_{BE1} - V_{BE2} = V_T \ln\!\left(\frac{I_{C1}}{I_{S1}}\right) - V_T \ln\!\left(\frac{I_{C2}}{I_{S2}}\right) \\ &= V_T \ln\!\left(\frac{I_{C1}/I_{S1}}{I_{C2}/I_{S2}}\right) = V_T \ln\!\left(\frac{I_{C1}}{I_{C2}} \cdot \frac{I_{S2}}{I_{S1}}\right). \end{aligned} \tag{5.15}$$

Using Eq. (5.14), the general result is

$$\Delta V_{BE} = V_T \ln\!\left[\frac{I_{C1}}{I_{C2}} \cdot n\right]. \tag{5.16}$$

If the branch currents differ by a ratio $m = I_{C1}/I_{C2}$, this becomes $V_T \ln(n\,m)$, the form used to reduce the influence of the op-amp offset.

Step 4: Equal-current bandgap core. In the core, op-amp feedback forces both branches to carry the same current, $I_{C1} = I_{C2}$. The current ratio in Eq. (5.16) becomes one:

$$\boxed{\Delta V_{BE} = V_T \ln n = \frac{kT}{q} \ln n} \tag{5.17}$$

Step 5: Temperature dependence. Since k, q and n are constants, $\Delta V_{BE} \propto T$, i.e. it is PTAT, with

$$\frac{\partial\, \Delta V_{BE}}{\partial T} = \frac{k}{q} \ln n. \tag{5.18}$$

For $n = 16$ this is $0.086\ \text{mV/K} \times \ln 16 \approx +0.24\ \text{mV/K}$.

Why the strong temperature dependence of I_S disappears. Both V_{BE1} and V_{BE2} contain $I_S(T) \propto T^{4+m}e^{-E_g/kT}$, which makes each of them strongly CTAT. In the difference, both junctions have the same $J_S(T)$, so it cancels exactly (Step 2), and only the temperature-independent geometric ratio n is left. This is the reason why ΔV_{BE} is a clean PTAT quantity while V_{BE} is not a clean CTAT quantity.

5.1.3 The PTAT current

When ΔV_{BE} is developed across the resistor R_1 (this happens in the core because $V_X = V_Y$),

$$I_{PTAT} = \frac{\Delta V_{BE}}{R_1} = \frac{V_T \ln n}{R_1} = \frac{kT}{qR_1} \ln n. \tag{5.19}$$

Thus $I_{PTAT} \propto T$, provided that R_1 is temperature independent or that its own temperature coefficient is compensated. In practice the resistor type (for example, a low-TC polysilicon resistor) matters, because the resistor slope adds to or subtracts from the PTAT slope.

Numerical example for this core. With $n = 16$ and $V_T \approx 25.85\,\text{mV}$ at $T = 300\,\text{K}$,

$$\Delta V_{BE} = 25.85\ \text{mV} \times \ln 16 \approx 71.7\ \text{mV}, \qquad I_{PTAT} = \frac{71.7\ \text{mV}}{2\ \text{k}\Omega} \approx 36\,\mu\text{A}, \tag{5.20}$$

which agrees with the chosen branch current of about $35\,\mu$A. As a second example, with an area ratio of 8 and $R_1 = 10\,\text{k}\Omega$: $\Delta V_{BE} = 25.85\ \text{mV} \times \ln 8 \approx 53.8\,\text{mV}$ and $I_{PTAT} \approx 5.38\,\mu\text{A}$.

Remark on geometric ratios. The ratio n is an *effective emitter-area* ratio. It is normally realised by connecting N identical unit transistors in parallel, which gives $n = N$ and the best matching. If instead the area ratio is set through device dimensions, the parameter N is replaced by the geometry ratio, e.g. $N \to (W/L)_2/(W/L)_1$, and

$$\Delta V_{BE} = \frac{kT}{q} \ln\!\left(\frac{(W/L)_2}{(W/L)_1}\right). \tag{5.21}$$

Here W/L denotes the emitter geometry and the equation is valid only if I_S scales with this area, which is less accurate than unit-device replication because of perimeter and edge effects.

5.1.4 Combining CTAT and PTAT for zero temperature coefficient

A reference voltage with zero first-order temperature coefficient is obtained as

$$V_{ref} = V_{BE} + K\,V_T \ln n, \qquad \frac{\partial V_{ref}}{\partial T} = \underbrace{-1.6\ \text{mV/K}}_{\text{CTAT}} + \underbrace{K\,\frac{k}{q}\ln n}_{\text{PTAT}} = 0. \tag{5.22}$$

This requires $K \ln n \approx 1.6/0.086 \approx 18.6$, and then $V_{ref} \approx 0.75\ \text{V} + 18.6 \times 26\ \text{mV} \approx 1.2\ \text{V}$. This is the well-known bandgap voltage, close to E_g/q of silicon. The gain K is set by a resistor ratio (R_2/R_1 in the voltage-mode form $V_{ref} = V_{BE} + \frac{R_2}{R_1}\frac{kT}{q}\ln n$), so that

$$\frac{R_2}{R_1} = \frac{-\partial V_{BE}/\partial T}{(k/q)\ln n} \approx \frac{1.6}{0.24} \approx 6.7 \quad (n = 16). \tag{5.23}$$

In the current-mode core the same role is played by R_L/R_1. In this design the CTAT part is converted to a current through R_2 instead of being added as a voltage, which allows the output to be scaled down to about 500 mV, as shown in the following sections.

5.1.5 Forcing equal currents with feedback

In the core of Figure 5.3(a), the op-amp A_1 senses nodes X and Y and drives the common gate node P of the PMOS transistors M_1 and M_2. Feedback drives $V_X \approx V_Y$ (the op-amp input difference becomes $V_{diff} \approx 0$). Since M_1 and M_2 have equal W/L and equal V_{GS}, they carry equal currents, $I_{D1} = I_{D2}$. Then

$$V_{BE1} = V_{BE2} + I_{D2}R_1 \implies |I_{D2}|\,R_1 = V_T \ln n. \tag{5.24}$$

The branch current is therefore PTAT, $I_D = V_T \ln n/R_1$, and it is *mirrored* to any number of additional branches by copies of M_1 and M_2.

5.1.6 Generating the reference

In the output branch of Figure 5.3(a) (extra PMOS M_3 and resistor R_L), the copied current is converted back to a voltage. When resistors R_2 and R_3 Figure 5.3(b) are placed across Q_1

and Q_2 so that each junction contributes a V_{BE}/R current as well, the output takes the form

$$V_{out} = R_L\left(\frac{|V_{BE}|}{R_2} + \frac{V_T \ln n}{R_1}\right), \tag{5.25}$$

i.e. a weighted sum of a CTAT and a PTAT term. Choosing R_L/R_1 and R_2 so that the two slopes cancel gives a first-order temperature-independent output (about 500 mV in this design). The ratio $R_L/R_1 = 0.14$ follows from a 72 mV drop across R_1 and a 500 mV output.

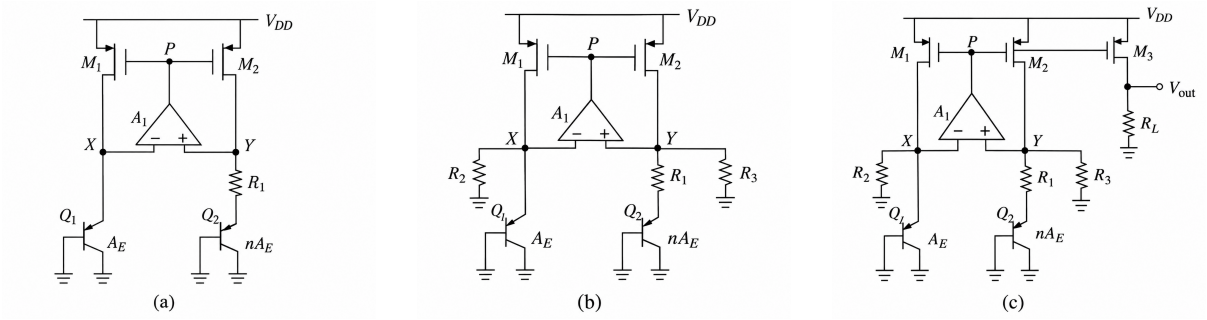


Figure 5.3: Schematic of the bandgap core: (a) the basic core with equal branch currents, (b) the addition of resistors R_2 and R_3 to create constant currents, and (c) the addition of an output branch (M_3, R_L). Redrawn after the structure in [1].

5.2 Topology Selection

Several topology decisions were made, each driven by the low supply voltage and by the requirements on accuracy and temperature stability.

- **Current-mode (PMOS-mirror) core instead of a voltage-mode core:** Mirroring a PTAT current into resistors lets the output be scaled to ≈ 500 mV, below the ≈ 1.2 V of a conventional bandgap. This is essential for a low-voltage supply, and the extra resistors R_2, R_3 and the output branch (M_3, R_L) give the designer freedom to set the output level.

- **Op-amp forcing $V_X = V_Y$ instead of a simple mirror:** The loop gives accurate equality of the two branch currents and a well-defined PTAT current that is not corrupted by channel-length modulation of the PMOS devices.

- **Bipolar devices Q_1, Q_2:** In a CMOS process, the vertical PNP (parasitic, collector to substrate) is available and is the only source of a V_{BE} with predictable temperature behaviour.

- **Large emitter-area ratio, small $|V_{BE}|$:** The limited voltage headroom makes it desirable to minimize $|V_{BE}|$, and hence to maximize the emitter area. However, the op-amp offset V_{OS} appears directly in the PTAT term, so n should also be large. A compromise is four unit transistors for Q_1 and 64 for Q_2 ($n = 16$), which balances area and offset sensitivity.

- **Ratio $m = 1$ (equal branch currents):** Scaling I_{D1} up by a factor m would reduce the effect of the offset as $|I_{D2}|R_1 = V_T \ln(n \cdot m)$, but it also raises $|V_{BE1}|$ (by $V_T \ln 16 \approx 66\,\text{mV}$ for $m = 16$ at 0°C) and worsens the MOS headroom at low temperature. Therefore $m = 1$ is kept and the low offset is obtained by proper op-amp design.

- **Start-up circuit:** The feedback loop has a degenerate stable state with zero current in all branches, so the core must incorporate a start-up circuit.

5.3 Main Components and Functionality

- **Bipolar Pair (Q_1 and Q_2):** They generate the CTAT voltage V_{BE} and, through their area ratio n, the PTAT voltage $\Delta V_{BE} = V_T \ln n$. Matching of the unit devices is essential.

- **Resistor R_1:** It converts ΔV_{BE} into the PTAT current $I_D = V_T \ln n / R_1$. Its value sets the current and the power consumption.

- **Op-Amp (A_1):** It senses V_X and V_Y and adjusts the gate voltage V_P so that $V_X = V_Y$. Its gain determines how accurately the two currents are equalised; its input offset directly degrades the PTAT term.

- **PMOS Current Sources (M_1 and M_2):** Their common gate node P is driven by the op-amp. They supply equal currents to the two bipolar branches and replicate the current in the output branch.

- **Resistors R_2 and R_3:** They provide the constant-current components V_{BE}/R that are added to the PTAT current in the current-mode topology, so that the output can be scaled below the bandgap voltage.

- **Output Branch (M_3 and R_L):** M_3 copies the current and R_L turns it into the reference voltage V_{out}.

Chain of Action:

1. Q_1 and Q_2 develop different V_{BE} values, since they have different emitter areas.
2. The op-amp senses the difference between V_X and V_Y and changes V_P.
3. M_1 and M_2 adjust their currents until $V_X = V_Y$.
4. The voltage across R_1 settles to $V_T \ln n$, which sets a PTAT current.
5. The current is mirrored by M_3 into R_L, and, together with the CTAT current through R_2, produces the temperature-compensated V_{out}.

5.4 Advantages and Limitations

Advantages:

- **Low-voltage operation:** The current-mode structure allows a reference of $\approx 500\,\text{mV}$, below the silicon bandgap voltage.
- **Accurate PTAT current:** Feedback forces equal branch currents independent of channel-length modulation.
- **Scalable output:** Resistor ratios (R_L/R_1, R_L/R_2) set the output level and the temperature coefficient.
- **Simple replication:** Additional currents are obtained by adding PMOS copies of M_1.

Limitations:

- **Op-amp offset:** The offset appears in the PTAT term and can only be reduced through n, m and good op-amp design.
- **Large area:** A large emitter-area ratio (here 64 units) needs significant silicon area.
- **Voltage headroom:** At low temperatures $|V_{BE}|$ is large and the MOS transistors have little headroom.
- **Start-up:** The zero-current state requires a start-up circuit.

Design Process Steps:

1. Q_1, Q_2: Select the emitter area and the ratio n for a compromise between $|V_{BE}|$, offset sensitivity and area.

2. I_D, R_1: Choose the branch current from the power budget and set $R_1 = V_T \ln n / I_D$.

3. M_1, M_2: Size W/L for matching, noise and output resistance, and check saturation over temperature.

4. R_2, R_3, R_L: Choose the ratios to cancel the CTAT and PTAT slopes and to set the output level.

5. Op-amp: Extract the common-mode range and output range from the core simulation, then design the amplifier (Chapter 4).

6. Complete core: Verify the temperature coefficient, power consumption and start-up over all specified temperatures.

Chapter 6

Amplifier Topologies

In the bandgap reference circuit, the operational amplifier plays a critical role in enforcing the necessary voltage constraints. For an ideal op-amp, it senses the voltage at node X (V_X) and node Y (V_Y) and amplifies this small difference ($V_{diff} = V_Y - V_X$). The amplifier changes its output voltage V_P, which is approximately $A(V_Y - V_X)$. The PMOS gate voltage in the core changes according to V_P to maintain the required operating conditions.

6.1 Five-Transistor OTA

The five-transistor Operational Transconductance Amplifier (OTA) is a compact and power-efficient topology commonly used in simple bandgap references.

Main Components and Functionality:

- **NMOS Differential Pair (M_X and M_Y):** The gate of M_X connects to input X, and the gate of M_Y connects to input Y. Their sources join at the common tail-current node, and their drains connect to the PMOS active-load network. They convert the voltage difference between X and Y into a difference in drain currents. Under ideal matched conditions where $V_X = V_Y$, the tail current is shared approximately equally ($I_{D,MX} \approx I_{D,MY} \approx 25\ \mu A$). Important parameters for this pair include transconductance (g_m), threshold voltage (V_{TH}), drain-source voltage (V_{DS}), and output conductance (g_{ds}).

- **PMOS Active-Load/Current-Mirror Pair (M_a and M_b):** These transistors act as the active load for M_X and M_Y. Their roles are to provide the load for the NMOS differential pair, convert the current imbalance into an output-voltage change, and help establish the OTA's voltage gain. The right-hand output branch connects to node P, which controls the gates of the core's PMOS transistors.

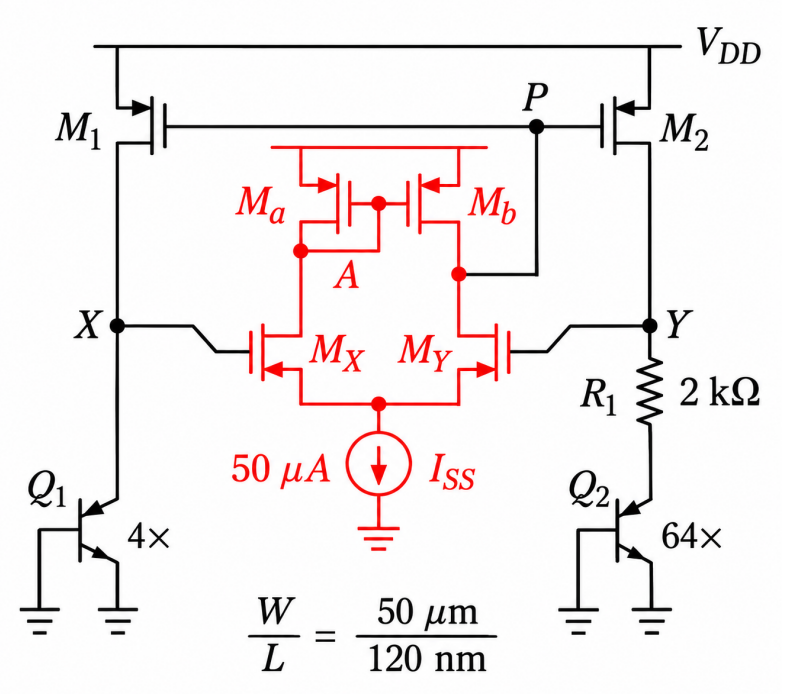


Figure 6.1: Schematic of the Five-Transistor OTA

- **Tail-Current Source (50 μA):** The tail-current source establishes the total current ($I_{D,MX} + I_{D,MY} \approx I_{SS}$). It affects the transconductance of the input pair, the amplifier's gain and speed, power consumption, and available voltage headroom.

Chain of Action:

1. M_X and M_Y sense V_X and V_Y.
2. Their drain currents change in response to the input voltage difference.
3. M_a and M_b convert this current imbalance into a change in voltage at node P.
4. The voltage at P changes the gate-source voltages of the core PMOS transistors (M_1 and M_2).
5. Their branch currents change to correct the initial voltage difference between X and Y.

Advantages:

- **Simple circuit:** Only five MOSFETs are required, making it relatively easy to design and integrate.

- **Low power consumption:** It uses a small bias current; the provided architecture uses just 50 μA.
- **Compact layout:** Fewer transistors generally require less silicon area and fewer internal nodes.
- **Differential input and active load:** It senses the voltage difference to support negative feedback and uses an active load to provide voltage gain without requiring large passive resistors.
- **Useful supply tracking:** The PMOS active load helps node P track supply variations, contributing to overall supply rejection.

Limitations:

- **Limited voltage gain:** Achieves a gain of approximately 20, which is significantly lower than that of a two-stage amplifier.
- **Limited PSRR:** Supply disturbances can still affect the circuit, particularly at low temperatures and higher frequencies.
- **Restricted input common-mode range:** The NMOS input pair requires sufficient voltage headroom to operate correctly.
- **Operating-region sensitivity:** At low temperatures, the input transistors may approach the triode region, substantially reducing amplifier gain.
- **Mismatch, offset, and swing limits:** Differences between transistor characteristics create input-referred offset, and the output voltage swing is restricted by the range allowed by the PMOS load and BGR transistors.

6.2 Two-Stage CMOS Op-Amp

For designs requiring tighter regulation and reduced input error, a two-stage CMOS op-amp can be utilized to significantly increase the feedback loop gain.

Main Components and Functionality:

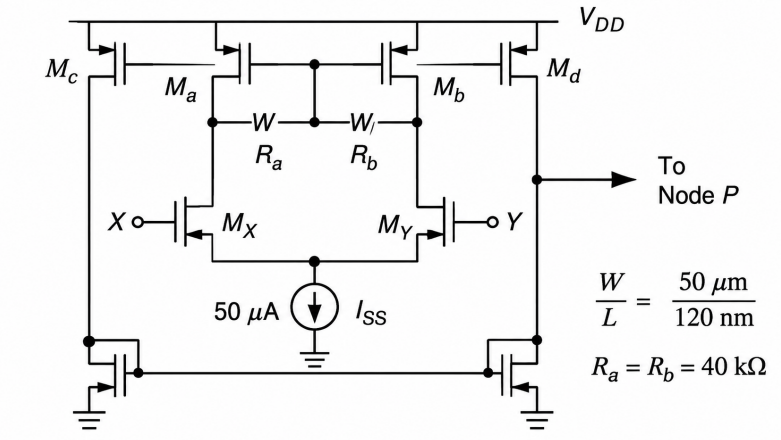


Figure 6.2: Schematic of the Two-Stage CMOS Op-Amp

- **NMOS Differential Input Pair (M_X and M_Y):** The gates connect to inputs X and Y, with sources joined at the tail-current branch and drains connected to the upper PMOS network. They convert a small input voltage difference into a difference between drain currents.

- **PMOS Active Loads (M_a and M_b):** Provide active loads for M_X and M_Y, convert the differential pair's current changes into voltage changes, and contribute to the first stage's voltage gain ($|A_{v1}| \approx g_m R_{out}$).

- **Bias-Setting Resistors (R_a and R_b):** These resistors help establish the output common-mode voltage of the first stage and a suitable DC operating point so the transistors can work in their intended regions. They are not the main gain-producing components.

- **Second-Stage and Bias Network (M_c and M_d):** M_c participates in establishing a reference branch current, while M_d provides the corresponding PMOS-side current for the output branch. This current-mirror arrangement establishes the required current available to the second gain stage before it amplifies the signal.

- **Current-Mirror Bias (Lower NMOS transistors):** A diode-connected NMOS develops a gate-source voltage corresponding to the current flowing through it. The mirror uses that voltage to establish a related current in the other branch, setting the second stage's bias current without requiring a separate ideal current source.

Advantages:

- **Higher Gain:** Provides significantly higher voltage gain than the five-transistor OTA.

- **Improved Supply Rejection:** Achieves better low-frequency PSRR within the BGR circuit.
- **Stable Biasing:** Bias currents are reliably established by transistor mirrors and resistors.
- **High Accuracy:** The high gain can effectively reduce the input error within the feedback loop.

Limitations:

- **Increased Complexity:** Requires more transistors and a much more complex bias network.
- **Higher Power:** Results in additional power consumption compared to single-stage designs.
- **Mismatch Sensitivity:** Highly sensitive to transistor mismatch and operating point variations.
- **Stability Challenges:** More internal nodes complicate frequency compensation and overall loop stability.

Design Process Steps:

1. M_X**,** M_Y**:** Select g_m/I_D and current density to meet the required input-pair transconductance.
2. M_a**,** M_b**:** Evaluate g_m/g_{ds}, current-mirror matching, and saturation headroom constraints.
3. M_c**,** M_d **and lower NMOS mirror:** Choose optimal dimensions for the required bias currents and output swing.
4. R_a**,** R_b**:** Verify the first-stage DC voltage and validate the second-stage bias point.
5. **Complete amplifier:** Thoroughly verify open-loop gain, output swing, power consumption, phase margin, and PSRR across all specified temperatures.

Chapter 7

Output Topologies

The output topology of a CMOS bandgap reference (BGR) determines how the temperature-compensated signal generated by the reference core is delivered as an output current or voltage. The selection of the output stage affects the accuracy, stability, and load-driving capability of the reference circuit.

Three output topologies are considered in this project: the current-output core, the voltage-output core, and the regulated-cascode voltage-output topology.

7.1 Current-Output Core

The current-output core combines the temperature-dependent effects of the proportional-to-absolute-temperature (PTAT) and complementary-to-absolute-temperature (CTAT) components to generate a branch current with reduced temperature dependence.

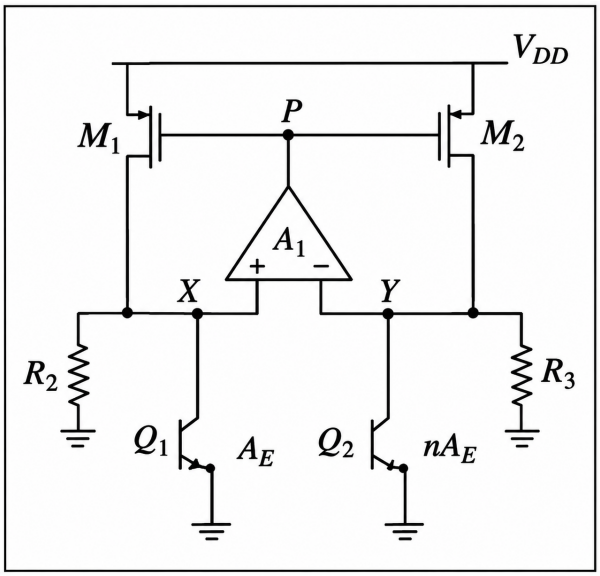


Figure 7.1: Current Output Core

Working Principle:

1. Generation of the base-emitter voltage difference: The BJT pair Q_1 and Q_2 has different effective emitter areas. This produces a difference in their base-emitter voltages, represented by ΔV_{BE}.

2. Generation of the PTAT current: The voltage difference is applied across resistor R_1, producing a current proportional to absolute temperature. The PTAT voltage is expressed as

$$\Delta V_{BE} = V_T \ln(n)$$

where

$$V_T = \frac{kT}{q}$$

Here, V_T is the thermal voltage, k is Boltzmann's constant, T is the absolute temperature in kelvin, q is the electronic charge, and n is the effective emitter-area or current-density ratio. The corresponding PTAT current is approximately

$$I_{\text{PTAT}} = \frac{\Delta V_{BE}}{R_1}$$

3. Introduction of the CTAT contribution: Resistors R_2 and R_3 introduce the CTAT contribution associated with the base-emitter voltage, V_{BE}. Since V_{BE} generally decreases as temperature increases, this contribution compensates for the temperature-dependent behavior of the PTAT component. 4. Combination of temperature-dependent currents: The PTAT and CTAT contributions are combined through appropriate circuit and resistor selection. The resistor values are adjusted to reduce the overall temperature dependence of the resulting branch current.

Advantages:

1. Provides a temperature-compensated reference current.
2. Combines PTAT and CTAT characteristics in a single reference core.
3. Can be used as the current-generating stage for subsequent output circuitry.

Limitations:

1. The compensation accuracy depends on resistor ratios and transistor characteristics.

2. Device mismatch and process variations can affect the output current.

3. Additional circuitry may be required when a stable output voltage is needed.

7.2 Voltage-Output Core

The voltage-output core converts the compensated current generated by the bandgap reference core into an output voltage. In the topology shown in the source, a PMOS transistor M_3 copies the reference current into an output branch containing a load resistor R_L.

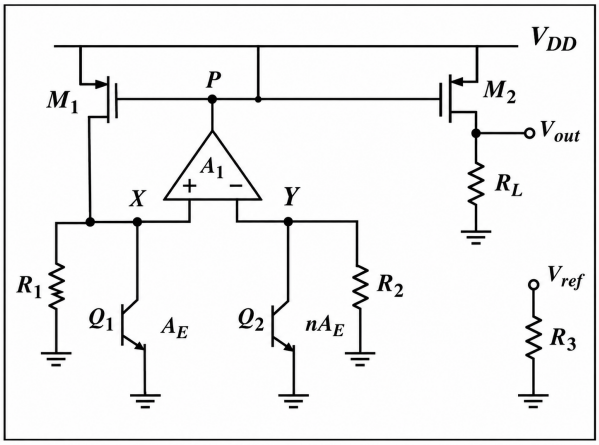


Figure 7.2: Voltage-Output Core

Working:

1. Generation of the compensated current: The BGR core combines the PTAT and CTAT contributions to establish a current with reduced temperature dependence.

2. Current mirroring: PMOS transistor M_3 copies the current established by the reference core into the output branch. Its gate is connected to the appropriate bias node of the PMOS current-mirror network.

3. Current flow through the load resistor: The mirrored current flows through the load resistor R_L, producing a voltage drop across it.

4. Conversion into an output voltage: The output voltage is determined by the current flowing through the output branch and the value of the load resistor. The output voltage can be expressed as

$$V_{\text{out}} = I_{M_3} R_L$$

where

- V_{out} is the output reference voltage.
- I_{M_3} is the current supplied by transistor M_3.
- R_L is the output load resistance.

  For a target output voltage of approximately 1.2 V, the load resistance can initially be estimated as

$$R_L = \frac{1.2\,\text{V}}{I_{M_3}}$$

  The actual resistance must be selected using the simulated output current and verified through circuit simulation.

Advantages:

1. Provides a direct voltage output from the compensated reference core.
2. Uses a relatively simple PMOS current mirror and resistor.
3. Allows the output voltage to be adjusted through the load resistance.
4. Is compatible with CMOS implementation using the SKY130 PDK.

Limitations:

1. Current-mirror mismatch can introduce output-voltage errors.
2. Variations in the reference current affect the output voltage.
3. The temperature coefficient of the load resistor can affect output stability.
4. The output voltage may vary with loading and transistor operating conditions.

7.3 Regulated-Cascode Voltage Output

The regulated-cascode voltage-output topology improves the accuracy of current copying by regulating the operating conditions of the output current-mirror branch. It uses an additional amplifier and a cascode transistor to reduce current-copying errors.

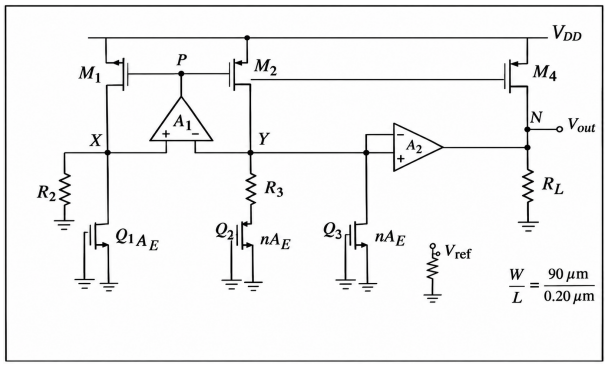


Figure 7.3: Regulated-Cascode Voltage Output

Circuit Components:

The topology shown in the source contains the following main components:

1. Main BGR core: Generates the temperature-compensated reference current.
2. Amplifier A_1: Regulates the core operating conditions so that the relevant internal voltages track one another.
3. Core current-setting transistor M_2: Establishes the current in the reference branch.
4. Output current-copying transistor M_3: Supplies the current to the output branch.
5. Cascode transistor M_4: Controls the operating conditions of the output branch.
6. Additional amplifier A_2: Controls the gate voltage of M_4.
7. Internal nodes V_N and V_Y: Represent the voltages compared by the regulation loop.
8. Load resistor R_L: Converts the output current into the output voltage V_{out}

7.4 Working

The regulated-cascode topology operates as follows:

1. Voltage sensing: Amplifier A_2 senses the internal voltages V_Y and V_N.
2. Voltage comparison: The amplifier compares these voltages and generates an error signal corresponding to their difference.

3. Gate-voltage adjustment: The output of A_2 adjusts the gate voltage of the cascode transistor M_4.

4. Control of transistor operation: The adjustment changes the operating point of M_4, driving the regulated internal voltage V_N toward the target voltage V_Y.

5. Reduction of current-copying error: Maintaining similar relevant drain-voltage conditions in the reference and output branches reduces the current-copying error between transistors M_2 and M_3.

6. Output-voltage generation: The current supplied by the output branch flows through R_L, generating the output voltage.

The regulation objective is

$$V_N \approx V_Y$$

This feedback condition helps maintain similar operating conditions in the relevant current-mirror devices and improves current-copying accuracy.

The output voltage is given by

$$V_{\text{out}} = I_{\text{out}} R_L$$

where I_{out} is the current delivered to the output branch.

Advantages:

1. Improves current-mirror accuracy by regulating internal node voltages.

2. Reduces errors caused by differences in transistor drain voltages.

3. Can improve the stability of the output current under varying operating conditions.

4. Provides better control of the output branch than a simple current mirror.

Limitations:

1. Requires additional transistors and an amplifier.

2. Increases circuit complexity and design effort.

3. Adds power consumption due to the additional amplifier and bias circuitry.

4. Requires careful verification of loop stability and transistor operating regions.

Chapter 8

Software Requirement

To complete the design and simulation of a Bandgap Reference (BGR) circuit, we are using a set of open-source tools: NGSPICE, Xschem, KLayout, and the SkyWater 130 nm PDK. Each tool provides essential capabilities that collectively support schematic design, circuit simulation, physical layout, and layout verification.

8.1 XSchem

Xschem is an open-source schematic capture tool designed with VLSI and mixed-signal design in mind, offering a straightforward drag-and-drop interface for creating, managing, and simulating complex circuits.

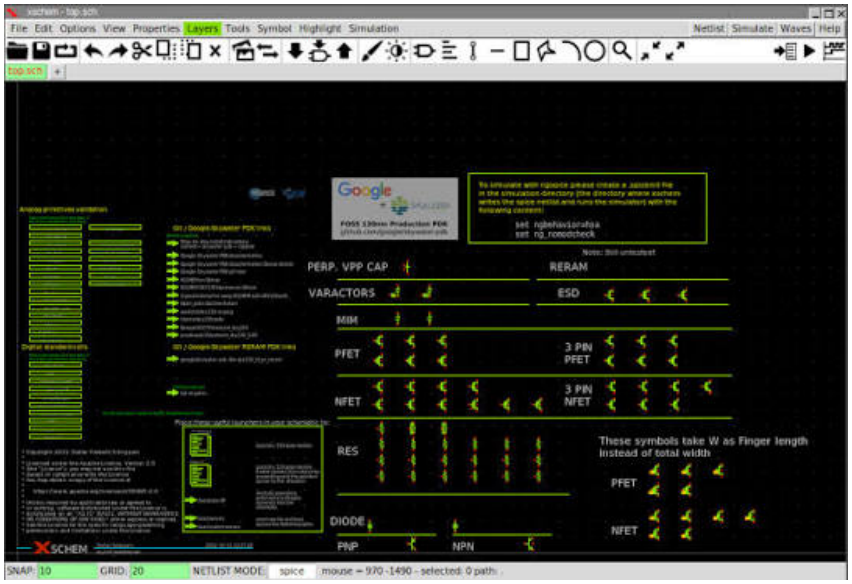


Figure 8.1: Xschem Home Page

Xschem supports hierarchical schematic design, enabling the proposed bandgap reference circuit to be organized into manageable functional blocks, while its extensive libraries and

custom component support facilitate the integration of specific transistor models and other circuit components required for accurate simulation. Xschem integrates seamlessly with NGSPICE, allowing us to simulate circuits directly from the schematic environment and to modify parameters without exporting separate netlists. This compatibility, combined with its scripting support for parametric sweeps, makes Xschem particularly effective for iterative design processes. Its intuitive interface also facilitates efficient debugging and layout preparation, making it accessible for both beginners and advanced users. As a highly customizable open-source tool, Xschem is well suited to support the detailed schematic requirements of this project.

8.2 NGSpice

NGSpice is a versatile open-source circuit simulator derived from SPICE3f5, supporting a wide range of analysis types, such as AC, DC, and transient analyses, making it suitable for analog and mixed-signal circuit design. With compatibility across various SPICE netlist formats, NGSpice enables accurate simulation and performance evaluation of the proposed bandgap reference circuit. It allows us to assess reference voltage stability, temperature dependence, power consumption, and response to supply variations, thereby helping to verify the circuit's performance against the desired design specifications.



Figure 8.2: NGSpice

Its capabilities include robust scripting and automation support, allowing us to perform parameter sweeps and batch processing, which are essential for optimizing transistor parameters and analyzing the performance of the bandgap reference circuit. Integration with Python and Xschem ensures a smooth workflow from schematic design to simulation, enabling iterative design adjustments without the need for intermediate file conversions. As an open-source

tool, NGSpice offers extensive post-processing options, including Fourier and noise analyses, which are useful for evaluating circuit behaviour under different operating conditions. These features make NGSpice well-suited for this project, where accurate modelling and simulation are essential for achieving the desired reference voltage stability, low power consumption, and temperature compensation.

8.3 KLayout

KLayout is an open-source IC layout design and visualization tool that provides essential features for physical design and verification of integrated circuits. It supports the creation and editing of custom layouts using multiple metal and device layers defined by the SkyWater 130 nm PDK. Its interactive layout editor enables precise placement of components, routing of interconnections, and visualization of different fabrication layers, making it suitable for analog IC design. KLayout supports design rule checking (DRC) and layout-versus-schematic (LVS) workflows through compatible verification tools and scripts, helping ensure that the layout satisfies fabrication requirements and corresponds to the intended circuit. Additionally, it supports layout inspection, GDSII file generation, and compatibility with other open-source EDA tools, facilitating integration into the overall IC design workflow. Its scripting and automation capabilities also help simplify repetitive layout tasks and improve design efficiency. These features, combined with its compatibility with the SkyWater 130 nm PDK, make KLayout a suitable tool for designing and verifying the physical layout of the proposed bandgap reference circuit.



Figure 8.3: KLayout

8.4 Skywater 130nm PDK

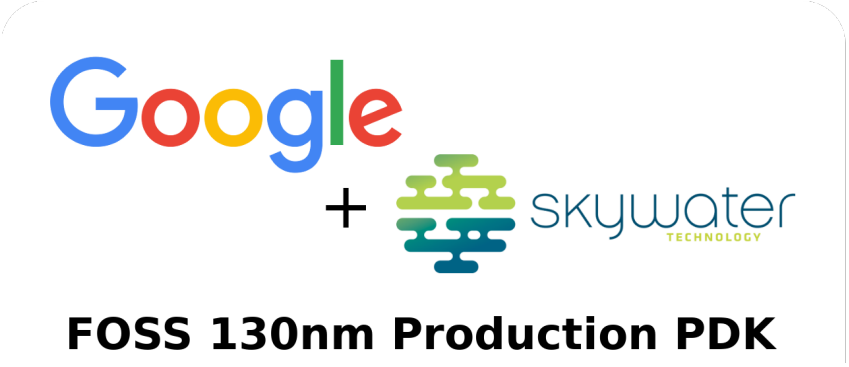


Figure 8.4: Skywater 130nm PDK

The SkyWater Open Source PDK, developed through a collaboration between Google and SkyWater Technology Foundry, offers a fully open-source Process Design Kit that supports realistic, manufacturable designs using SkyWater's 130 nm CMOS technology. Although it is primarily intended for research, education, and open-source IC development, it provides an excellent foundation for creating test chips and performing initial design verification. The SkyWater 130 nm PDK includes SPICE models for key components such as MOSFETs, capacitors, and resistors, which reflect fabrication characteristics and enable reliable NGSPICE simulations for evaluating the performance of the proposed bandgap reference circuit. This PDK also provides detailed design rules tailored to the 130 nm process, ensuring that layouts created using tools like KLayout can be checked against manufacturing constraints. Additionally, it provides essential device models and parasitic information for evaluating circuit behaviour under different operating conditions. By providing open-source access, the SkyWater PDK facilitates educational use and practical IC design experience, making it suitable for this project's objectives. It supports an integrated design workflow encompassing schematic design, circuit simulation, physical layout, and verification, promoting accessible and hands-on learning in analog IC design.

Chapter 9

Design Specifications

9.1 Introduction

The design specifications define the target electrical and operational characteristics of the proposed CMOS Bandgap Reference (BGR) implemented using the SKY130 PDK. The specifications are selected to achieve a stable reference voltage with low power consumption, reliable startup, and operation over a wide temperature range. These requirements serve as the basis for circuit design, transistor sizing, simulation, layout, and post-layout verification.

9.2 Target Design Specifications

The target specifications of the proposed bandgap reference circuit are summarized in Table 9.1.

9.3 Justification of Design Parameters

9.3.1 Supply Voltage

A supply voltage of 1.8 V is selected for the proposed SKY130 CMOS implementation. It provides the operating supply for the reference core and associated biasing circuitry. The transistor operating points and voltage headroom must be verified to ensure correct operation at this supply voltage.

Table 9.1: Target Design Specifications

Parameter	**Target Value**	**Design Justification**
Supply Voltage	1.8 V	Compatible with the nominal core supply used for the SKY130 CMOS implementation.
Reference Voltage	Approximately 1.2 V	Provides a stable voltage reference based on the combination of CTAT and PTAT voltage components.
Operating Temperature Range	−40 °C to 125 °C	Evaluates reference stability across a wide temperature range relevant to practical IC applications.
Power Consumption	$< 100\,\mu$W	Limits the power dissipation and supports energy-efficient analog circuit operation.
Startup Time	$< 10\,\mu$s	Aims to establish the reference voltage quickly after power is applied, subject to startup circuit and load conditions.

9.3.2 Reference Voltage

The target reference voltage is approximately 1.2 V. A conventional bandgap reference combines a complementary-to-absolute-temperature (CTAT) voltage, typically derived from a bipolar junction, with a proportional-to-absolute-temperature (PTAT) voltage.

The output can be expressed as

$$V_{\mathrm{REF}} = V_{\mathrm{CTAT}} + KV_{\mathrm{PTAT}}, \tag{9.1}$$

where K is a scaling factor determined by the circuit design. Proper scaling reduces the first-order temperature dependence of the output voltage.

9.3.3 Operating Temperature Range

The target operating range is from −40 °C to 125 °C. Over this range, the temperature coefficients of the CTAT and PTAT components must compensate for each other as closely as possible. DC operating-point and temperature-sweep simulations are required to evaluate the

resulting reference-voltage variation.

9.3.4 Power Consumption

The power consumption is targeted to remain below $100\,\mu$W. This requirement limits the current drawn by the reference core and its biasing circuitry. The total power is evaluated using

$$P_{\mathrm{DC}} = V_{\mathrm{DD}} I_{\mathrm{DD}}, \tag{9.2}$$

where V_{DD} is the supply voltage and I_{DD} is the total supply current.

At a supply voltage of 1.8 V, the corresponding current limit is

$$I_{\mathrm{DD}} < \frac{100\,\mu\mathrm{W}}{1.8\,\mathrm{V}} \approx 55.6\,\mu\mathrm{A}. \tag{9.3}$$

Therefore, the total supply current must remain below approximately $55.6\,\mu$A to satisfy the specified power target.

9.3.5 Startup Time

The startup time is targeted to be less than $10\,\mu$s. A bandgap reference may have a zero-current operating condition in which the circuit fails to establish its intended bias point automatically. A startup circuit helps move the core into its correct operating state after power is applied. Transient simulation is required to verify startup behavior and measure the time taken for the reference output to settle within a specified tolerance of its final value.

9.4 Specification Verification

The design targets are evaluated through the following verification procedures:

- **DC operating-point analysis:** Verify the reference output voltage, bias currents, and transistor operating regions.
- **Temperature-sweep analysis:** Evaluate the reference voltage over the specified temperature range and calculate its temperature coefficient.
- **Power analysis:** Determine the total supply current and calculate the corresponding power consumption.

- **Transient analysis:** Verify reliable startup and measure the settling time after power is applied.
- **Post-layout verification:** Repeat the relevant simulations using extracted parasitics to assess the effects of layout on circuit performance.

The above specifications provide the design objectives for the proposed CMOS bandgap reference. Their achievement must be established through circuit simulation and post-layout verification.

Chapter 10

Design Methodology

10.1 Theoretical Background: The g_m/I_D Methodology

The traditional textbook approach to analog circuit design relies heavily on the square-law equation for MOS transistors, where drain current is approximated as $I_D = \frac{1}{2}\mu C_{ox}(W/L)V_{ov}^2$ and transconductance is given by $g_m = 2I_D/V_{ov}$. While this provides a foundational intuition where the overdrive voltage (V_{ov}) acts as the primary design knob, the square-law model is only valid for long-channel devices operating heavily in strong inversion.

In modern short-channel devices, the square-law estimations deviate significantly from actual silicon behavior. As technology scales, devices experience velocity saturation and mobility degradation at short channel lengths (L). Furthermore, analog design often pushes transistors to operate across weak, moderate, and strong inversion regions, where V_{ov} loses its theoretical meaning, particularly near and below the threshold voltage (V_T). Consequently, hand calculations based on the square law can sit far from the highly accurate BSIM SPICE simulation results.

To resolve this discrepancy, the g_m/I_D methodology was adopted. The primary goal of this model-based methodology is to keep the insight of hand analysis but feed it simulated device data instead of the square-law equations. By utilizing a set of width-independent metrics extracted directly from the technology library (such as SKY 130), designers can systematically size transistors with high accuracy.

10.1.1 Regions of Inversion

The inversion level of a transistor, rather than its absolute width (W) or length (L) alone, fundamentally dictates its behavior. The g_m/I_D methodology categorizes operation into three

primary regions:

- **Weak Inversion ($g_m/I_D \gtrsim 20\,\mathbf{V}^{-1}$):** Current flows primarily via diffusion and has an exponential relationship with V_{GS}. This region offers the highest transconductance efficiency and lowest power consumption. However, it comes at the cost of the lowest transit frequency (f_T) and requires large device widths and capacitances.

- **Moderate Inversion ($g_m/I_D \approx 10$ to $20\,\mathbf{V}^{-1}$):** Current is a mix of both drift and diffusion mechanisms[cite: 22]. This region provides an optimal balance between transconductance efficiency and operational speed, making it typical for Operational Transconductance Amplifier (OTA) input pairs and gain devices. Because there is no simple hand equation for this region, lookup tables and charts are essential.

- **Strong Inversion ($g_m/I_D \lesssim 10\,\mathbf{V}^{-1}$):** Current is dominated by drift, following a square-law trend. This region yields high speed (high f_T) and small device sizes. However, it is the least efficient region, requiring the most current per unit of g_m, and it demands higher voltage headroom.

10.2 Device Characterization and Plot Analysis

To accurately design the analog blocks, it is essential to first characterize the standard NMOS (`nfet_01v8`) and PMOS (`pfet_01v8`) devices available in the technology node. The parameter extraction is achieved by sweeping the gate-to-source voltage (V_{GS} for NMOS, V_{SG} for PMOS) across various channel lengths while fixing the width, and measuring the corresponding small-signal parameters.

10.2.1 NMOS Device Characterization (`nfet_01v8`)

From this testbench, we extract four primary width-independent design charts. The g_m/I_D metric acts as the central reference axis for navigating these design trade-offs.

The exact same principles apply to the PMOS devices. The setup below extracts the analogous characteristics by sweeping the source-to-gate voltage (V_{SG}) while keeping the width constant and stepping the channel length.

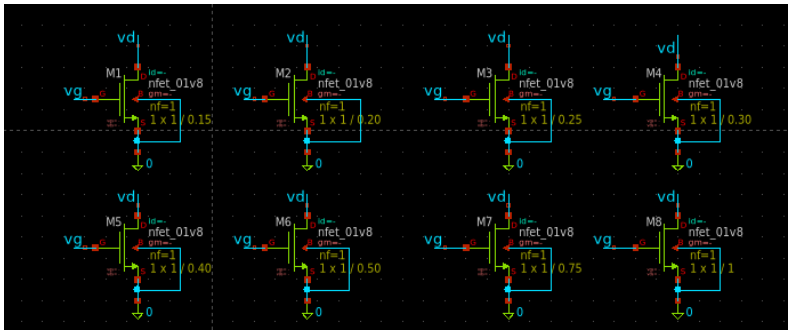


Figure 10.1: Circuit simulation setup for NMOS (`nfet_01v8`) characterization

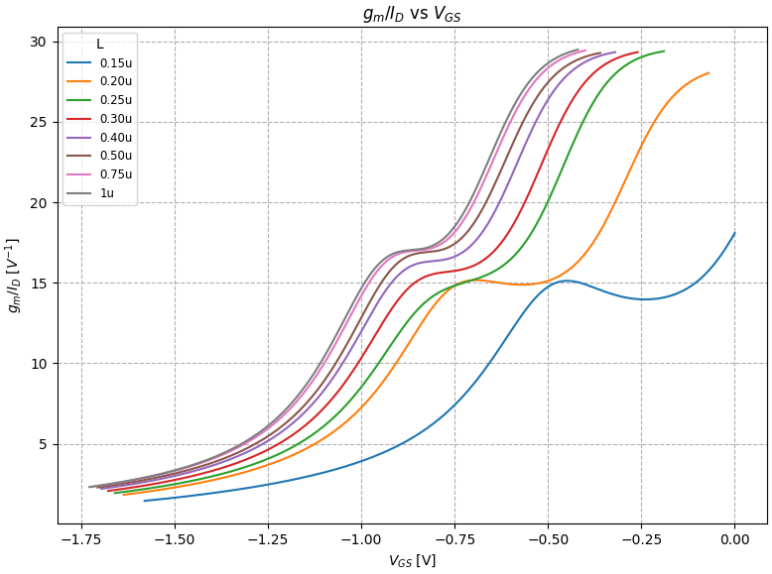


Figure 10.2: NMOS: Transconductance Efficiency (g_m/I_D) vs V_{GS}

Plot Description: Figure 10.2 maps the transconductance efficiency to the bias voltage. The g_m/I_D ratio indicates how much transconductance is generated for a given unit of DC bias current; essentially, it dictates the current needed for the circuit ($I_D = g_m \div (g_m/I_D)$). Higher values correspond to weak inversion, while lower values push the device into strong inversion.

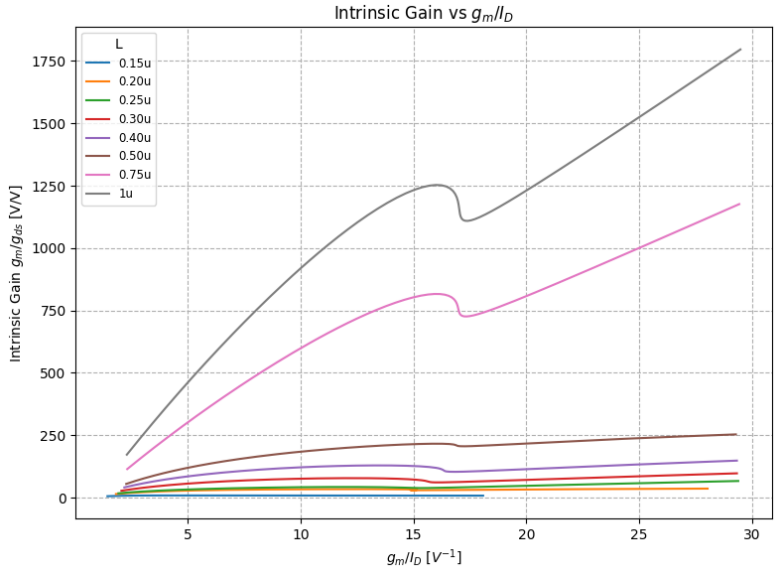


Figure 10.3: NMOS: Intrinsic Gain (g_m/g_{ds}) vs g_m/I_D

Plot Description: Figure 10.3 illustrates the intrinsic gain, which establishes the absolute maximum bound for single-stage voltage gain. As seen in the plot, higher intrinsic gain is achieved at higher g_m/I_D values (lower V_{ov}) and grows significantly as the channel length (L) increases. Designers use this chart to select an appropriate channel length to meet the gain specifications of an amplifier.

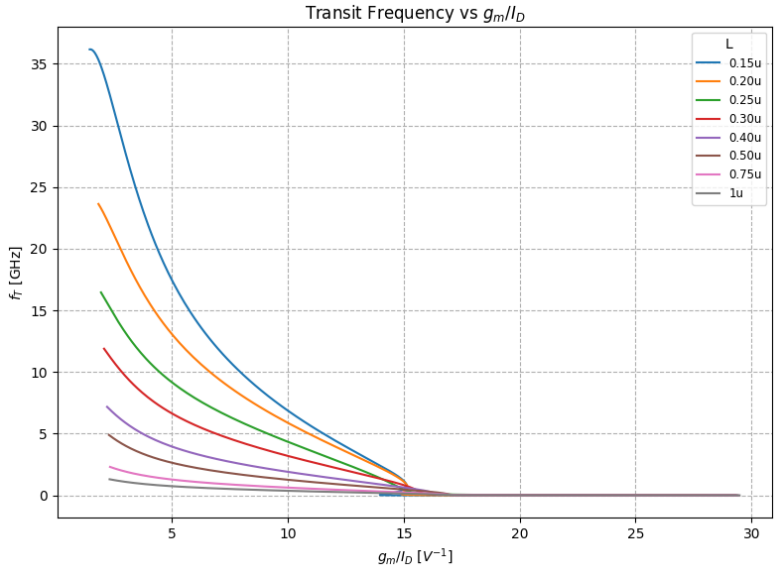


Figure 10.4: NMOS: Transit Frequency (f_T) vs g_m/I_D

Plot Description: Figure 10.4 plots the transit frequency, defined as $f_T \approx g_m/(2\pi C_{gg})$, which represents the speed limit of the transistor. The chart demonstrates a fundamental trade-off in analog design: while a transistor is a more efficient transconductor at low V_{ov} (high g_m/I_D), the transit frequency falls as g_m/I_D rises. Strong inversion yields the fastest operation, while weak inversion is the slowest.

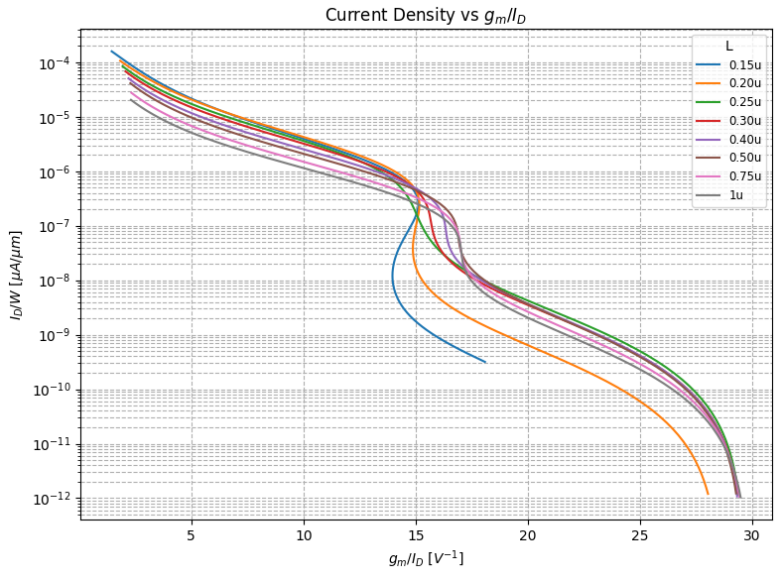


Figure 10.5: NMOS: Current Density (I_D/W) vs g_m/I_D

Plot Description: Figure 10.5 provides the current density (I_D/W). Once the required g_m/I_D is chosen to balance gain and speed, and the total drain current (I_D) is calculated, this chart directly tells the designer how to size the device. The final transistor width is determined by $W = I_D \div (I_D/W)$.

10.2.2 PMOS Device Characterization (`pfet_01v8`)

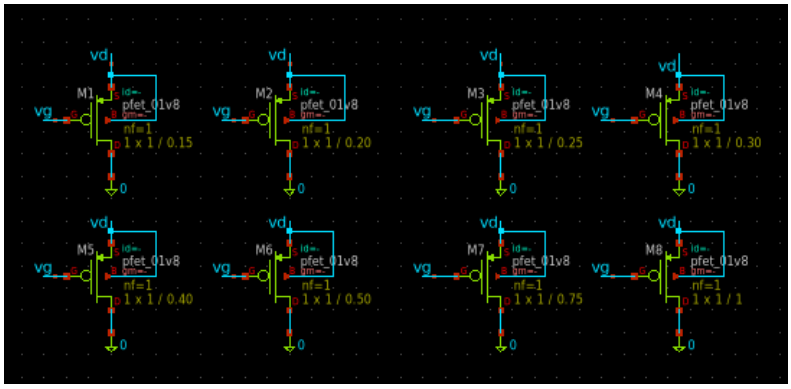


Figure 10.6: Circuit simulation setup for PMOS (`pfet_01v8`) characterization

By utilizing these comprehensive lookup charts, the sizing of analog components (such as differential pairs, current mirrors, and active loads) becomes a deterministic process mapping directly to the specifications of gain, bandwidth, and power consumption, bypassing the inaccuracies of traditional square-law hand calculations.

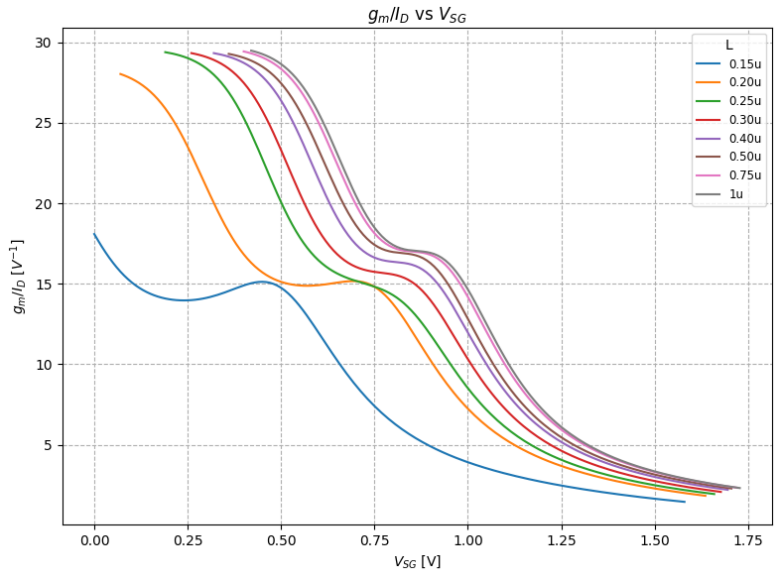


Figure 10.7: PMOS: Transconductance Efficiency (g_m/I_D) vs V_{SG}

Plot Description: Figure 10.7 illustrates how the transconductance efficiency of the PMOS device varies with the source-to-gate voltage (V_{SG}). Similar to the NMOS, higher g_m/I_D values correspond to the weak inversion region (high efficiency, low speed), while lower values push the PMOS into strong inversion.

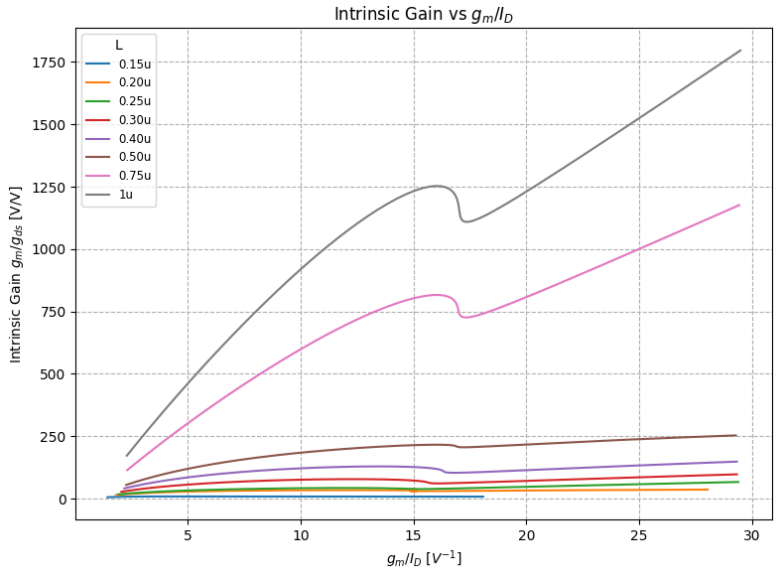


Figure 10.8: PMOS: Intrinsic Gain (g_m/g_{ds}) vs g_m/I_D

Plot Description: Figure 10.8 plots the intrinsic gain of the PMOS device against its transconductance efficiency. The chart shows that intrinsic gain increases significantly as the channel length (L) is increased. Designers use this plot to determine the minimum channel length required to achieve a target voltage gain for PMOS active loads or amplifier stages.

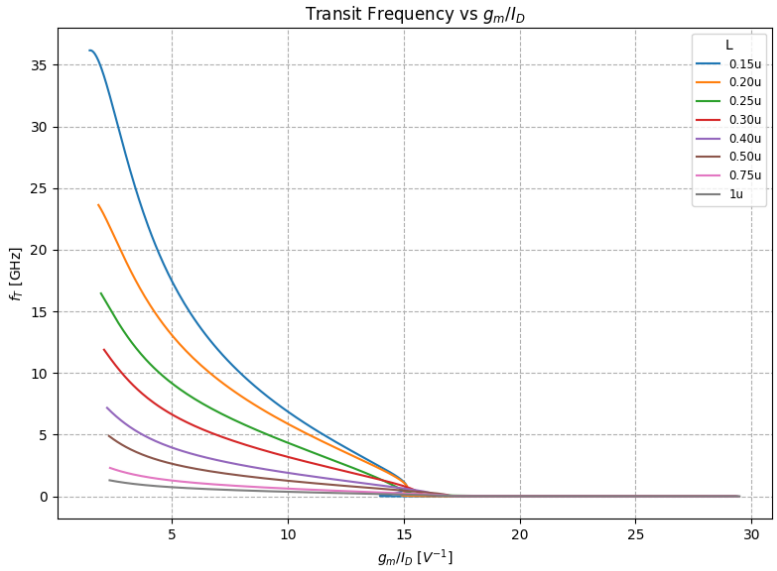


Figure 10.9: PMOS: Transit Frequency (f_T) vs g_m/I_D

Plot Description: Figure 10.9 displays the transit frequency (f_T), representing the operational speed limit of the PMOS transistor. The trade-off between power efficiency and speed is clearly visible: as g_m/I_D increases (moving towards weak inversion for better efficiency), the transit frequency drops rapidly. Strong inversion must be used if high-frequency operation is required.

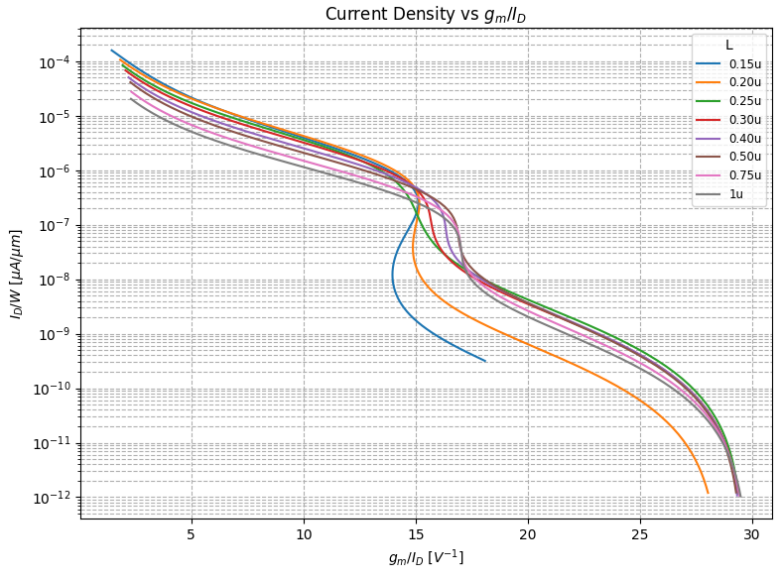


Figure 10.10: PMOS: Current Density (I_D/W) vs g_m/I_D

Plot Description: Figure 10.10 provides the current density (I_D/W) for the PMOS device. This chart is the final step in the sizing methodology. Once the required g_m/I_D is selected from the previous charts to balance gain and bandwidth, this plot yields the exact current density. The final PMOS width is then calculated by dividing the known branch current by this density ($W = I_D \div (I_D/W)$).

Chapter 11

Ideal Bandgap Reference Circuit Design

11.1 Circuit Topology

The bandgap reference circuit is designed using SKY130 130-nm CMOS technology with a supply voltage of 1.8 V and a target reference voltage of 1.2 V. The circuit consists of a BJT temperature-sensing core, a five-transistor operational transconductance amplifier (OTA), MOSFET current mirrors, resistors, and an output branch.

Table 11.1: Initial Design Specifications

Parameter	Value
Technology	SKY130, 130-nm CMOS
Supply voltage	1.8 V
Target reference voltage	1.2 V
Temperature range	−40°C to 125°C
Nominal temperature	27°C
Target power	Below 100 μW
BJT emitter-area ratio	1 : 16
OTA topology	Five-transistor OTA

Figure 11.1 shows the bandgap reference circuit. The BJT core generates the CTAT and PTAT components, while the MOSFET network establishes the bias currents. The resistors determine the scaling of the PTAT voltage, and the output branch provides the reference voltage.Since ideal device models were used in the initial simulation, a separate startup circuit was not incorporated into the design, as the idealized conditions simplified the establishment of the intended operating point.

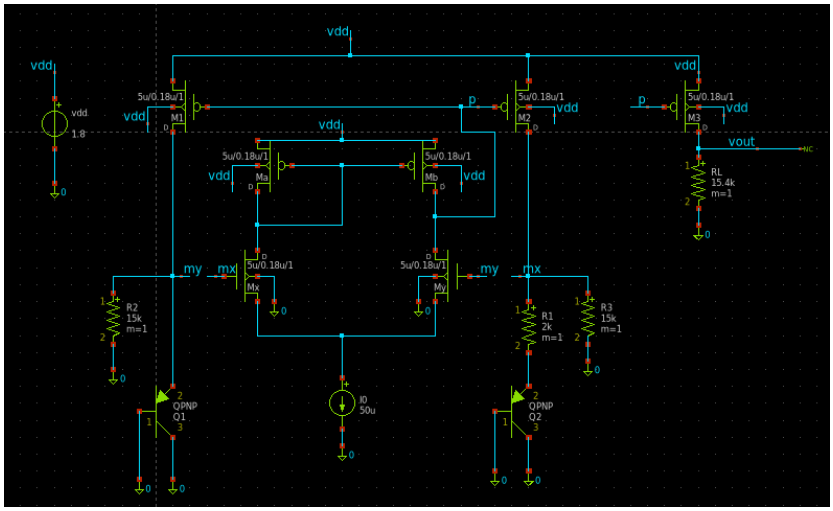


Figure 11.1: Schematic of the ideal bandgap reference circuit.

11.2 Design of the PTAT Voltage

11.2.1 Thermal Voltage Calculation

The thermal voltage is expressed as

$$V_T = \frac{kT}{q}, \tag{11.1}$$

where $k = 1.380649 \times 10^{-23}$ J/K is Boltzmann's constant, $q = 1.602176634 \times 10^{-19}$ C is the elementary charge, and T is the absolute temperature.

At the nominal temperature of 27°C,

$$T = 27 + 273.15 = 300.15\,\text{K}. \tag{11.2}$$

Substituting the values,

$$V_T = \frac{1.380649 \times 10^{-23} \times 300.15}{1.602176634 \times 10^{-19}}. \tag{11.3}$$

Therefore,

$$\boxed{V_T = 25.86\,\text{mV}}. \tag{11.4}$$

11.2.2 Difference Between Base-Emitter Voltages

The PTAT voltage is generated using two BJTs operating at different current densities. Its magnitude is

$$\Delta V_{BE} = \frac{kT}{q} \ln(N), \tag{11.5}$$

where N is the emitter-area ratio when the collector currents are equal.

For the selected ratio,

$$N = \frac{A_2}{A_1} = 16. \tag{11.6}$$

Substituting,

$$\Delta V_{BE} = 25.86\,\text{mV} \times \ln(16). \tag{11.7}$$

Since $\ln(16) = 2.7726$,

$$\boxed{\Delta V_{BE} = 71.72\,\text{mV}}. \tag{11.8}$$

This calculation assumes the required current-density ratio is maintained in the actual circuit.

11.2.3 Calculation of the PTAT Current

The PTAT current is obtained by applying ΔV_{BE} across the resistor R_1:

$$I_{\text{PTAT}} = \frac{\Delta V_{BE}}{R_1}. \tag{11.9}$$

Selecting $R_1 = 2\,\text{k}\Omega$,

$$I_{\text{PTAT}} = \frac{71.72 \times 10^{-3}}{2 \times 10^{3}}. \tag{11.10}$$

Therefore,

$$\boxed{I_{\text{PTAT}} = 35.86\,\mu\text{A}}. \tag{11.11}$$

11.2.4 PTAT Scaling Resistor

The PTAT contribution required for a 1.2 V reference is initially estimated using $V_{BE} = 0.662\,\text{V}$.

$$V_{\text{PTAT}} = V_{\text{REF}} - V_{BE}. \tag{11.12}$$

Substituting,

$$V_{\text{PTAT}} = 1.2 - 0.662 = 0.538\,\text{V}. \tag{11.13}$$

The required scaling resistor is

$$R_2 = \frac{V_{\text{PTAT}}}{I_{\text{PTAT}}}. \tag{11.14}$$

Therefore,

$$R_2 = \frac{0.538}{35.86 \times 10^{-6}}, \tag{11.15}$$

$$\boxed{R_2 \approx 15.0\,\text{k}\Omega}. \tag{11.16}$$

The scaling factor is

$$K = \frac{R_2}{R_1} = \frac{15\,\text{k}\Omega}{2\,\text{k}\Omega}. \tag{11.17}$$

Hence,

$$\boxed{K = 7.5}. \tag{11.18}$$

The assumed V_{BE} is an initial design estimate. The final resistor ratio must be adjusted using the actual BJT operating-point voltage and temperature characteristics.

11.3 Calculation of the Reference Voltage

The first-order bandgap reference equation is

$$V_{\text{REF}} = V_{BE} + K\Delta V_{BE}. \tag{11.19}$$

Using $K = 7.5$ and $\Delta V_{BE} = 71.72\,\text{mV}$,

$$V_{\text{PTAT}} = 7.5 \times 71.72\,\text{mV}. \tag{11.20}$$

Therefore,

$$\boxed{V_{\text{PTAT}} = 0.538\,\text{V}}. \tag{11.21}$$

Using the initial $V_{BE} = 0.662\,\text{V}$,

$$V_{\text{REF}} = 0.662 + 0.538. \tag{11.22}$$

Thus,

$$\boxed{V_{\text{REF}} \approx 1.200\,\text{V}}. \tag{11.23}$$

For a measured V_{BE}, the required scaling factor can be calculated directly from

$$K = \frac{V_{\text{REF,target}} - V_{BE}}{\Delta V_{BE}}. \tag{11.24}$$

For example, if the measured $V_{BE} = 0.680\,\text{V}$,

$$K = \frac{1.2 - 0.680}{0.07172}, \tag{11.25}$$

$$\boxed{K \approx 7.25}. \tag{11.26}$$

This value illustrates the adjustment required when the actual BJT voltage differs from the initial estimate.

11.4 Five-Transistor OTA Design

The five-transistor OTA consists of two differential input MOSFETs, two current-mirror load MOSFETs, and one tail-current transistor. Its low transistor count makes it suitable for

compact analog circuits.

The differential input voltage is

$$V_{\text{ID}} = V_{+} - V_{-}. \tag{11.27}$$

The small-signal output is approximated by

$$v_o = A_{\text{OL}} v_{\text{ID}}, \tag{11.28}$$

where A_{OL} is the open-loop voltage gain.

11.4.1 Tail Current and Branch Currents

The initial tail current is selected as

$$I_{\text{tail}} = 50\,\mu\text{A}. \tag{11.29}$$

For equal differential inputs,

$$I_{D1} = I_{D2} = \frac{I_{\text{tail}}}{2}. \tag{11.30}$$

Substituting,

$$I_{D1} = I_{D2} = \frac{50}{2}\,\mu\text{A}, \tag{11.31}$$

$$\boxed{I_{D1} = I_{D2} = 25\,\mu\text{A}}. \tag{11.32}$$

11.4.2 Transconductance

For a MOSFET operating in strong-inversion saturation,

$$I_D = \frac{1}{2}\mu C_{\text{ox}} \frac{W}{L} V_{\text{OV}}^2, \tag{11.33}$$

where $V_{\text{OV}} = V_{GS} - V_{\text{TH}}$.

The transconductance is

$$g_m = \frac{2I_D}{V_{\mathrm{OV}}}. \tag{11.34}$$

For an illustrative overdrive voltage of 0.20 V,

$$g_m = \frac{2 \times 25\,\mu\mathrm{A}}{0.20\,\mathrm{V}}, \tag{11.35}$$

$$\boxed{g_m = 250\,\mu\mathrm{S}}. \tag{11.36}$$

11.4.3 MOSFET Aspect Ratio

Rearranging the drain-current equation,

$$\frac{W}{L} = \frac{2I_D}{\mu C_{\mathrm{ox}} V_{\mathrm{OV}}^2}. \tag{11.37}$$

For illustration only, assume an effective process parameter of $200\,\mu\mathrm{A/V^2}$.

$$\frac{W}{L} = \frac{2 \times 25\,\mu\mathrm{A}}{200\,\mu\mathrm{A/V^2} \times (0.20)^2}. \tag{11.38}$$

Therefore,

$$\boxed{\frac{W}{L} = 6.25}. \tag{11.39}$$

This illustrative process parameter is not asserted to be the actual SKY130 value. Final transistor dimensions must be calculated from the selected PDK model and verified by simulation.

For equal mirror currents, the initial mirror-sizing relationship is

$$\frac{I_{D2}}{I_{D1}} \approx \frac{(W/L)_2}{(W/L)_1} = 1. \tag{11.40}$$

Hence,

$$\boxed{(W/L)_2 = (W/L)_1}. \tag{11.41}$$

11.4.4 OTA Voltage Gain

The approximate open-loop gain is

$$A_{\mathrm{OL}} \approx g_m R_{\mathrm{out}}. \tag{11.42}$$

For an illustrative output resistance of 100 kΩ,

$$A_{\mathrm{OL}} = 250 \times 10^{-6} \times 100 \times 10^{3}, \tag{11.43}$$

$$\boxed{A_{\mathrm{OL}} = 25\,\mathrm{V/V}}. \tag{11.44}$$

The gain in decibels is

$$A_{\mathrm{OL,dB}} = 20\log_{10}(25), \tag{11.45}$$

$$\boxed{A_{\mathrm{OL,dB}} \approx 28\,\mathrm{dB}}. \tag{11.46}$$

The five-transistor OTA offers compactness, low transistor count, and simple differential amplification. Its limitations include finite gain, limited output swing, and restricted drive capability. The actual gain and stability must be verified in simulation.

11.5 Resistor and Output-Load Design

The initial resistor values are

$$R_1 = 2\,\mathrm{k\Omega}, \quad R_2 = 15\,\mathrm{k\Omega}, \quad R_3 = 15\,\mathrm{k\Omega}. \tag{11.47}$$

The voltage across R_1 is

$$V_{R_1} = I_{\mathrm{PTAT}} R_1, \tag{11.48}$$

$$V_{R_1} = 35.86\,\mu\mathrm{A} \times 2\,\mathrm{k\Omega}, \tag{11.49}$$

$$\boxed{V_{R_1} = 71.72\,\text{mV}}. \tag{11.50}$$

The voltage across R_2 is

$$V_{R_2} = I_{\text{PTAT}} R_2, \tag{11.51}$$

$$V_{R_2} = 35.86\,\mu\text{A} \times 15\,\text{k}\Omega, \tag{11.52}$$

$$\boxed{V_{R_2} = 0.538\,\text{V}}. \tag{11.53}$$

If R_3 has a voltage drop of 0.538 V, its current is

$$I_{R_3} = \frac{0.538}{15 \times 10^3}, \tag{11.54}$$

$$\boxed{I_{R_3} = 35.87\,\mu\text{A}}. \tag{11.55}$$

The actual current depends on the voltage across R_3 in the final circuit.

For an output load of $R_L = 15.4\,\text{k}\Omega$,

$$I_L = \frac{V_{\text{REF}}}{R_L}. \tag{11.56}$$

Substituting $V_{\text{REF}} = 1.2\,\text{V}$,

$$I_L = \frac{1.2}{15.4 \times 10^3}, \tag{11.57}$$

$$\boxed{I_L = 77.92\,\mu\text{A}}. \tag{11.58}$$

The load power is

$$P_L = \frac{V_{\text{REF}}^2}{R_L}, \tag{11.59}$$

$$P_L = \frac{1.2^2}{15.4 \times 10^3}, \tag{11.60}$$

$$\boxed{P_L = 93.51\,\mu\text{W}}. \tag{11.61}$$

The load alone consumes approximately $93.51\,\mu$W. Consequently, the circuit cannot satisfy a strict total power limit of $100\,\mu$W with this load while also consuming appreciable internal bias power.

11.6 Temperature Compensation

The temperature coefficient of the reference is

$$\frac{dV_{\text{REF}}}{dT} = \frac{dV_{BE}}{dT} + K\frac{d(\Delta V_{BE})}{dT}. \tag{11.62}$$

For first-order temperature compensation,

$$\frac{dV_{\text{REF}}}{dT} \approx 0. \tag{11.63}$$

Thus,

$$K = -\frac{dV_{BE}/dT}{d(\Delta V_{BE})/dT}. \tag{11.64}$$

For $N = 16$,

$$\frac{d(\Delta V_{BE})}{dT} = \frac{k}{q}\ln(16). \tag{11.65}$$

Using $k/q = 86.17\,\mu\text{V/K}$,

$$\frac{d(\Delta V_{BE})}{dT} = 86.17 \times 2.7726\,\mu\text{V/K}, \tag{11.66}$$

$$\boxed{\frac{d(\Delta V_{BE})}{dT} = 238.9\,\mu\text{V/K}}. \tag{11.67}$$

For an illustrative CTAT coefficient of $-1.8\,\text{mV/K}$,

$$K = \frac{1.8}{0.2389}, \tag{11.68}$$

$$\boxed{K \approx 7.53}. \tag{11.69}$$

This is close to the initial resistor ratio of 7.5. The actual CTAT coefficient must be extracted from the BJT simulation before finalizing the resistor ratio.

11.7 Simulation Results

11.7.1 PTAT and CTAT Characteristics

Figure 11.2 shows the temperature dependence of the PTAT and CTAT components. The CTAT component decreases with temperature, whereas the PTAT component increases. Their opposing temperature coefficients reduce the variation of the reference voltage.

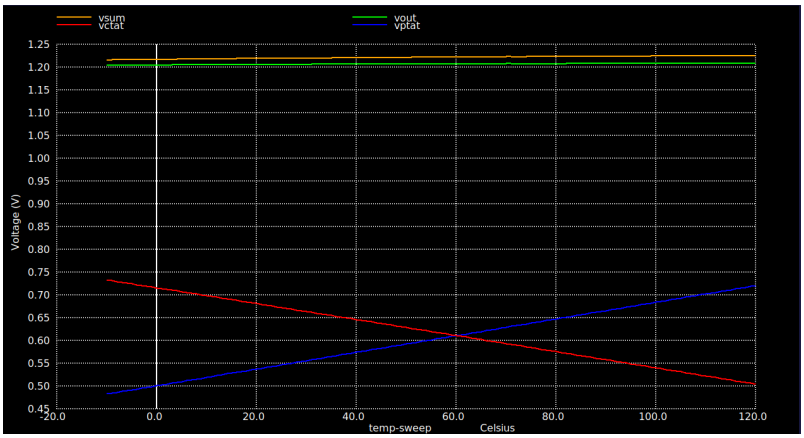


Figure 11.2: Simulated PTAT and CTAT voltage characteristics.

11.7.2 Reference Voltage Versus Temperature

Figure 11.3 presents the reference voltage over temperature.

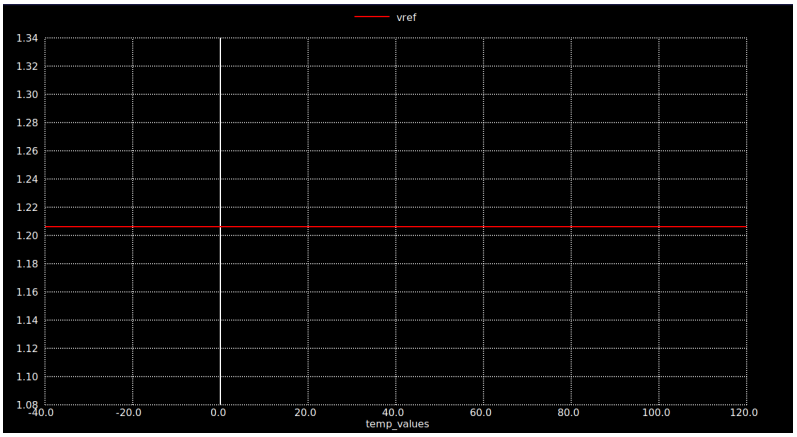


Figure 11.3: Simulated reference voltage versus temperature.

The displayed graph indicates an approximately constant output near 1.205 V. The exact minimum and maximum voltages must be extracted from simulation data to calculate the temperature coefficient accurately.

11.7.3 Transient Response

The transient output is evaluated by plotting the reference voltage against time after the supply is applied.

For a 1% settling criterion,

$$|V_{\mathrm{REF}}(t) - V_{\mathrm{REF,final}}| \leq 0.01 V_{\mathrm{REF,final}}. \tag{11.70}$$

For a final output of 1.2 V,

$$\Delta V_{\mathrm{settle}} = 0.01 \times 1.2 = 12\,\mathrm{mV}. \tag{11.71}$$

Therefore, the output must remain within

$$\boxed{1.188\,\mathrm{V} \leq V_{\mathrm{REF}} \leq 1.212\,\mathrm{V}}. \tag{11.72}$$

The startup time must be measured from the actual transient waveform.

11.8 Power Consumption

The total power consumption is

$$P_{\text{total}} = V_{DD} I_{DD}. \tag{11.73}$$

For $V_{DD} = 1.8\,\text{V}$ and a total supply current of $50\,\mu\text{A}$,

$$P_{\text{total}} = 1.8 \times 50\,\mu\text{A}, \tag{11.74}$$

$$\boxed{P_{\text{total}} = 90\,\mu\text{W}}. \tag{11.75}$$

This result applies only if $50\,\mu\text{A}$ is the total current drawn from the supply. If it is only the OTA tail current, the complete circuit power will be different.

For a $100\,\mu\text{W}$ power limit,

$$I_{DD} < \frac{100\,\mu\text{W}}{1.8\,\text{V}}, \tag{11.76}$$

$$\boxed{I_{DD} < 55.56\,\mu\text{A}}. \tag{11.77}$$

11.9 Conclusion

The bandgap reference design was analysed using the PTAT–CTAT compensation principle. At 27°C, the calculated thermal voltage is 25.86 mV, and the base-emitter voltage difference for an emitter-area ratio of 16 is approximately 71.72 mV.

With $R_1 = 2\,\text{k}\Omega$, the calculated PTAT current is $35.86\,\mu\text{A}$. A scaling resistor of $15\,\text{k}\Omega$ produces a PTAT contribution of approximately 0.538 V. Combined with the assumed $V_{BE} = 0.662\,\text{V}$, the calculated reference voltage is 1.2 V.

The five-transistor OTA provides compact differential amplification, while the MOSFET current mirrors establish the bias currents. The displayed simulation indicates an output near 1.205 V. Final verification requires the actual SKY130 device models, complete operating-point analysis, temperature sweeps, transient simulation, and physical verification.

Chapter 12

Conclusion

The project presents the preliminary design of a CMOS bandgap reference (BGR) circuit using the open-source SKY130 130 nm PDK. By combining proportional-to-absolute-temperature (PTAT) and complementary-to-absolute-temperature (CTAT) components, the project explores the generation of a stable reference voltage of approximately 1.2 V from a 1.8 V supply. The theoretical analysis and preliminary NGSPICE simulations indicate an output voltage close to the target value of 1.205 V. The study of transistor characteristics using the (g_m/I_D) methodology also provides a systematic approach to transistor sizing and device analysis for analog circuit design.

Challenges remain in establishing the circuit's overall performance under variations in temperature and supply voltage, as well as in evaluating power consumption and startup behaviour. Since Phase I is limited to theoretical analysis and preliminary simulation, further verification is required to assess the reliability and stability of the proposed design. Future work will focus on refining the circuit, implementing its physical layout using KLayout, and performing design rule checking (DRC), layout-versus-schematic (LVS) verification, and post-layout simulation to evaluate layout-induced parasitic effects. This project thus establishes a foundation for further development of a reliable CMOS bandgap reference, with potential applications in power management, data conversion, and other analog and mixed-signal integrated circuits.

References

[1] B. Razavi, "The design of a low-voltage bandgap reference," *IEEE Solid-State Circuits Magazine*, vol. 13, no. 3, pp. 6–13, Summer 2021.

[2] K. M. Cañonero, J. S. Bernales, A. G. Macapundag, S. A. Khalid, and S. E. Maestre, "Design of a bandgap voltage reference for an 8-bit SAR-ADC for powerline monitoring using SKY130 PDK," in *Proc. 10th Int. Conf. Integrated Circuits and Microsystems (ICICM)*, 2025.

[3] M. Gunawan, G. C. M. Meijer, J. Fonderie, and J. H. Huijsing, "A curvature-corrected low-voltage bandgap reference," *IEEE Journal of Solid-State Circuits*, vol. 28, no. 6, pp. 667–670, Jun. 1993.

[4] Y. Osaki, T. Hirose, N. Kuroki, and M. Numa, "1.2-V supply, 100-nW, 1.09-V bandgap and 0.7-V supply, 52.5-nW, 0.55-V subbandgap reference circuits for nanowatt CMOS LSIs," *IEEE Journal of Solid-State Circuits*, vol. 48, no. 6, pp. 1530–1540, Jun. 2013.

[5] G. Zhu, X. Huang, Y. You, Y. Li, W. Guo, and Z. Zhu, "Multi-cell battery sensing and protection IC with integrated low-temperature-drift reference for series battery pack management," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 72, no. 10, pp. 5634–5645, Oct. 2025.

[6] Y.-Y. Huang, M. Takamiya, and P.-H. Chen, "Serial-stacked single-inductor dual-source triple-output energy harvesting interface with 2D-MPPT for battery-free IoT devices," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 72, no. 11, pp. 6592–6603, Nov. 2025.

[7] K. Chen, L. Petruzzi, R. Hulfachor, and M. Onabajo, "A 1.16-V 5.8-to-13.5-ppm/°C curvature-compensated CMOS bandgap reference circuit with a shared offset-cancellation

method for internal amplifiers," *IEEE Journal of Solid-State Circuits*, vol. 56, no. 1, pp. 267–276, Jan. 2021.

[8] X. Ming, L. Hu, Y.-L. Xin, X. Zhang, D. Gao, and B. Zhang, "A high-precision resistor-less CMOS compensated bandgap reference based on successive voltage-step compensation," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 65, no. 12, pp. 4086–4097, Dec. 2018.

[9] B. L. Hunter and W. E. Matthews, "A ±3 ppm/°C single-trim switched capacitor bandgap reference for battery monitoring applications," *IEEE Transactions on Circuits and Systems I: Regular Papers*, vol. 64, no. 4, pp. 777–786, Apr. 2017.